

# Presolicitation Notice

**SUBJECT** Fort Snelling National Cemetery – National Shrine Project 894-NSP25-002  
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## GENERAL INFORMATION

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|---------------------------------------|--------------------------------------------------------------------------|
| <b>CONTRACTING OFFICE'S ZIP CODE*</b> | 68116                                                                    |
| <b>SOLICITATION NUMBER*</b>           | 36C78626B0009                                                            |
| <b>RESPONSE DATE/TIME/ZONE</b>        | 04-14-2026 15:00 CENTRAL TIME, CHICAGO, USA                              |
| <b>ARCHIVE</b>                        | 30 DAYS AFTER THE RESPONSE DATE                                          |
| <b>SET-ASIDE</b>                      | SDVOSBC                                                                  |
| <b>PRODUCT SERVICE CODE*</b>          | Z1QA                                                                     |
| <b>NAICS CODE*</b>                    | 561730                                                                   |
| <b>PLACE OF PERFORMANCE</b>           | Fort Snelling National Cemetery<br>7601 34th Ave S<br><br>Minneapolis MN |
| <b>POSTAL CODE</b>                    | 55450                                                                    |
| <b>COUNTRY</b>                        | USA                                                                      |

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## CONTACT INFORMATION

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|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>CONTRACTING OFFICE ADDRESS</b> | Department of Veterans Affairs<br>National Cemetery Administration<br>Contracting Services<br>18434 Joplin Road<br>Triangle VA 22172 |
| <b>POINT OF CONTACT*</b>          |                                                                                                                                      |

\*=Required Field

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## ADDITIONAL INFORMATION

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|---------------------------------------|---------------------------------------------------------------|
| <b>AGENCY'S URL</b>                   | <a href="https://www.cem.va.gov/">https://www.cem.va.gov/</a> |
| <b>URL DESCRIPTION</b>                | National Cemetery Administration                              |
| <b>AGENCY CONTACT'S EMAIL ADDRESS</b> |                                                               |
| <b>EMAIL DESCRIPTION</b>              |                                                               |

# Presolicitation Notice

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## DESCRIPTION

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### **Fort Snelling National Cemetery – National Shrine Project**

**Project No.:** 894-NSP25-002

**Set-Aside:** 100% Service-Disabled Veteran-Owned Small Business (SDVOSB)

**Place of Performance:** Fort Snelling National Cemetery, 7601 34th Ave S, Minneapolis, MN 55450

#### **1. Notice**

This is a Presolicitation Notice to inform industry of the Department of Veterans Affairs (VA), National Cemetery Administration's intent to issue a solicitation for the Fort Snelling National Cemetery National Shrine Project. The acquisition is anticipated to be set aside 100% for SDVOSB firms.

This notice is issued for planning purposes only. It is not a request for proposals, bids, or quotes. No pricing information shall be submitted at this time.

#### **2. Project Overview**

The Contractor gaining award must provide all personnel, management, supplies, transportation, equipment and supervision necessary to perform and complete all work to provide complete section renovation for the cemetery sections indicated that includes, but not limited to: site surveying; preparing grading plans and headstone realignment plans; vegetation removal, grading and drainage, excavation and backfill; headstone identification/tagging; headstone handling; flat marker geogrid support system; headstone re-installation; headstone cleaning; turf renovation; and site restoration work. Note: "Headstone" includes both upright headstones and flat markers.

The Contractor will complete all work within **Three Hundred and Sixty-Five (365)** calendar days after receipt of Notice of Award, subject to all terms, conditions, provisions, and schedules of the contract. Performance must be in accordance with the requirements, conditions, terms and representations contained in this solicitation.

#### **3. NAICS Information**

The Government anticipates use of the following NAICS code:

- NAICS 561730 – Landscaping Services

#### **4. SDVOSB Verification**

Only SDVOSB firms verified through SBA's Veteran Small Business Certification (VetCert) at both time of offer and award will be eligible to compete. Firms will be required to submit a UEI, CAGE code, and evidence of VetCert approval.

#### **5. Submission Instructions**

This presolicitation notice does not require a response. A separate Sources Sought / RFI has been issued for capability statements from SDVOSB firms.

When the solicitation is released, it will include complete instructions for submission of offers.

# Presolicitation Notice

## 6. Disclaimer

This announcement is not a request for proposals and does not constitute a commitment by the Government to proceed with a solicitation or award. The Government is not responsible for any costs incurred as a result of this notice. Any future solicitation will be posted at SAM.gov.

End of Document

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### C. LIST OF EXHIBITS

### D. LIST OF ATTACHMENTS

## **A. STATEMENT OF WORK**

### **GENERAL REQUIREMENTS**

#### **1.1 SUMMARY OF WORK:**

- A. **WORK AREA:** All work is to be performed on Fort Snelling National Cemetery, in the burial sections and surrounding areas listed in the work area schedule. (See Exhibit 'A' – Fort Snelling National Cemetery Renovation Areas.)
- B. **CEMETERY SECTION RENOVATION:** The Contractor gaining award must provide all personnel, management, supplies, transportation, equipment and supervision necessary to perform and complete all work to provide complete section renovation for the cemetery sections indicated that includes, but not limited to: site surveying; preparing grading plans and headstone realignment plans; vegetation removal, grading and drainage, excavation and backfill; headstone identification/tagging; headstone handling; flat marker geogrid support system; headstone re-installation; headstone cleaning; turf renovation; and site restoration work. Note: "Headstone" includes both upright headstones and flat markers.

The Contractor will complete all work within **Three Hundred and Sixty-Five (365)** calendar days after receipt of Notice of Award, subject to all terms, conditions, provisions, and schedules of the contract. Performance must be in accordance with the requirements, conditions, terms and representations contained in this solicitation.

**Note:** The effective period of Turf Renovation component of the project shall occur between April 15 to June 30 for spring, and from August 15 to October 15 for fall of awarded period of performance. Turf must be established prior to the onset of winter season. Bare ground is not acceptable during this time.

- C. One of the most important considerations in erecting or emplacing a headstone or marker is that of precision in alignment. Much of the beauty that exists or should exist in a national cemetery is the direct result of perfectly aligned headstones and markers. It only takes one headstone or marker out of alignment to spoil the appearance of an entire row, and thereby, the appearance of the entire section. The appearance of the finished work shall be gravesite section(s) of contoured grass landscape, with headstones set and aligned with each other in the vertical, horizontal, lateral and diagonal directions within the tolerances specified herein. It is critical that existing conditions are surveyed, and contours adjusted as necessary and approved by the COR to produce such an appearance. Failure to achieve this finished work will be basis for accepting or not accepting the finished work.

- D. Provide all work to complete a single gravesite and the immediate surrounding turf in accordance with the statement of work, drawings and specifications. Immediate surrounding turf is that turf that begins at the gravesite and runs up to the adjoining gravesites, section boundaries, fences, curbs, sidewalks, garden or tree lines and as further defined on the drawings. Award will be made based on entire cemetery Sections listed including "Row or "Rows" of gravesites and may contain Government Upright Headstones and/or Flat Markers as defined herein. A single "Row" of gravesites is defined as an array of continuous gravesites that begins and ends at the limits of a single section. The number of actual gravesites in each row may and does vary.
- E. The quantities of gravesites listed for each section is for bidding purposes only. Prior to beginning work of gravesite renovation, the Contractor must provide a survey of the area within the scope of work by a professional land surveyor licensed in the state of Minnesota . The survey will locate the physical boundaries of the limit of work shown on the Drawings, gravesites markers, (upright or flat), reserved gravesites, section markers, control monuments, edge of pavement, curb lines, walkways, trees, valve boxes, vaults, walls, fences, and all other visible structures within the limit-of-work. The survey will provide a summary table listing total quantities of each type of gravesite marker (upright and flat markers) within each burial section and total area of turf renovation. Quantities listed must be certified by the professional land surveyor. (See Section "SITE SURVEY" for additional requirements.) Based on the survey, the actual quantities of gravesites and area of turf renovation will be determined. If there are inconsistencies in the number of gravesites and turf renovation, the CO will negotiate with the contractor via a Request for Equitable Adjustment based off the original bid price.
- F. **TO ENSURE THE ACCURACY OF MARKER PLACEMENT DURING THE RAISE AND REALIGNMENT PROJECT, THE FOLLOWING PROCEDURES WILL BE FOLLOWED BY CEMETERY PERSONNEL / CONTRACTING OFFICERS REPRESENTATIVE (COR) AND THE CONTRACTOR BEFORE THE RAISE / REALIGNMENT PROCESS BEGINS:**

**The Contracting Officer Representative (COR) and the Contractor or designee as listed in the official contract documents must conduct a joint gravesite verification and inspection prior to the removal of headstones/markers from any gravesite. The COR will provide a copy of the Burial Register Report from the Burial Operation Support System (BOSS) to the Contractor. The Burial Register Report will be used to verify the section, row, grave number, first and last name of each decedent and the condition of the headstone/marker included in the scope of work (SOW). All inconsistencies will be researched, discussed and resolved with the Cemetery Director prior to moving any headstone or marker from the gravesite. Upon completion of the joint survey and corrective actions as necessary, the COR and the Contractor must certify the verification and inspection, by signing the Burial**

**Register Report printout. The CO/COR will give the approval to begin the project. When the Contractor finishes working each day, a joint re-inspection will be completed for all headstones and markers raised & realigned. The Contractor and the COR will re-inspect and verify the accuracy of the placement of headstones or markers on the correct gravesite using the same Burial Register Report that was used for the initial inspection. A copy of the original, signed verification and inspection form must be forwarded to the Contracting Officer, a copy of the report will be maintained by the Cemetery. Copies of the inspections/verification forms will be provided to the contractor's designee and COR for their official contract files.**

**The Contractor will protect all headstones and markers from damage during the process. The Contractor will ensure that headstones and marker are protected from damage or being discolored from dirt or soil contact. Contractor must submit transmittals to the COR on the method of elevating headstones and markers for approval.**

**HEADSTONES AND MARKERS WILL NOT BE PHYSICALLY REMOVED FROM THE GRAVESITE DURING THE PROJECT.**

**PRICE/COST SCHEDULE**

| ITEM NO.                                 | DESCRIPTION                                                                                                | UNIT     | TOTAL PRICE |
|------------------------------------------|------------------------------------------------------------------------------------------------------------|----------|-------------|
| BASE BID PERIOD OF PERFORMANCE: 365 DAYS |                                                                                                            |          |             |
| 001                                      | <b><i><u>BASE BID:</u></i></b><br><b><i>FULL RENOVATION</i></b><br><b><i>(SECTIONS: 30, 31, 33, 2)</i></b> | LUMP SUM | \$ _____    |
|                                          |                                                                                                            |          |             |
|                                          |                                                                                                            |          |             |

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| <b>TOTAL OF BASE BID</b>       |                 |
|--------------------------------|-----------------|
| BASE BID – LUMP SUM BID AMOUNT | \$ _____        |
|                                |                 |
|                                |                 |
|                                |                 |
| <b>GRAND TOTAL BID AMOUNT</b>  | <b>\$ _____</b> |

Note (1): The professional land surveyor must measure the total square feet of area to receive turf replacement/renovation to include all of the cemetery sections defined in the work area. Excluded from the total square-footage calculation for turf replacement/renovation are the footprint of paving, gravesite markers, headstones, section markers, valve boxes, trees, mulched beds, and any other site feature which deducts from the area of turf.

Note (2): Irrigation Systems: Irrigation systems are to be protected in place. Any damage to the irrigation system must be repaired by the Contractor at no additional cost to the Government. Work to be accomplished under this contract does not involve new installation of irrigations lines, sprinkler heads or other associated fixtures. However, the work does involve the raising and/or lowering of sprinkler heads, and irrigation valve boxes. The Contractor must submit a request to the COR/Cemetery Director to make temporary adjustments to the irrigation system in the sections subject to the contract so as not to interfere with the Contractor's activities. Temporary adjustments include: temporary removal and capping of sprinkler heads; temporary adjustment of arc patterns of sprinkler heads; and temporary adjustments to irrigation scheduling. All irrigation adjustments must be restored to their original condition prior to installation of new sod.

## 1.2 BURIAL SECTION DATA:

| SNELLING (894) FY26 SHRINE PROJECT                                                                                                                                                           |                |                        |                |                      |                  |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------|----------------|----------------------|------------------|-----------|
| Cemetery Section                                                                                                                                                                             | Gravesite Type | Approx. Gravesite Size | Headstone Type | Number of Headstones | Approx Turf Area | Work Type |
| <b>Full Section Renovation</b> as described herein includes all the following: Headstone Support System, Marker Raise, Reset, and Realign, Complete Turf Renovation, and Headstone Cleaning. |                |                        |                |                      |                  |           |
| <b>Raise &amp; Realign</b> as described herein includes all the following: Headstone Support System, Marker Raise, Reset, and Realign, Turf Renovation                                       |                |                        |                |                      |                  |           |
| BASE BID                                                                                                                                                                                     |                |                        |                |                      |                  |           |
| 30                                                                                                                                                                                           | CRYPT          | 3X8                    | UPRIGHT MARBLE | 3589                 | 95383            | FULL      |
| 31                                                                                                                                                                                           | CRYPT          | 3X8                    | UPRIGHT MARBLE | 3928                 | 104544           | FULL      |
| 33                                                                                                                                                                                           | CREMAINS       | 4X4                    | UPRIGHT MARBLE | 5415                 | 95304            | FULL      |
| 2                                                                                                                                                                                            | CASKET         | 6X10                   | UPRIGHT MARBLE | 736                  | 48576            | FULL      |

**1.3 THE NCA MISSION:** The National Cemetery Administration honors veterans with a final resting place and lasting memorials that commemorate their service to our Nation. National cemeteries are national shrines. The standards of maintenance, appearance and operational procedures performed by the Contractor at this cemetery must reflect this nation's concern for those interred there. For this reason, the Contractor's strict adherence to the specifications is required and is essential. The Contractor must demonstrate a clear understanding of, and the sensitivity to, environmental issues during the performance of the contract.

## 1.4 **CONTRACTING OFFICER'S REPRESENTATIVE (COR):**

- A. The VA Contact Person is Marcus Syverson the Fort Snelling National Cemetery Assistant Director and COR for the project. The COR or their designee(s) will make site visits on a regular basis. The COR may designate an "authorized on site designee" to observe work progress.
- B. The Contracting Officer's Representative (COR) responsibilities will include, but not be limited to surveillance of services required under this contract; determining the adequacy of Contractor performance according to the specifications of this contract; acting as the Government's representative; ensuring compliance with contract requirements insofar as the work is concerned; and advising the CO of any contract issues. The COR cannot authorize deviations from the specifications and requirements of this contract. In the event the Contractor deviates without approval from the CO, such deviation shall be at the risk of the Contractor and any cost related thereto shall be borne by the Contractor. The COR designated for this proposed contract will be officially appointed in writing by the CO.
- C. If work is authorized to be performed after hours or on weekends/holidays and an emergency should occur, the Contractor shall contact the VA Police Office in the absence of the COR or their designee. The VA Police will then contact the COR or take appropriate action.

**1.5 CEMETERY LOCATION:** The services will be performed at one (1) location:

**FORT SELLING NATIONAL CEMETERY**  
7601 34th Ave S, Minneapolis, MN 55450

**1.6 CONTRACTOR DUTIES:**

- A. Contractor Superintendent: A competent and experienced Contractor Superintendent must be provided by the Contractor for not less than two (2) hours a day whenever work is being performed - other than trash and debris pick-up. The Contractor Superintendent must have not less than three (3) years experience as a direct supervisor of a commercial service operation that included surveying, soil excavation, and site construction in industrial, commercial or public sites. The Contractor Superintendent must ensure all specifications are being met, ensure contract work does not conflict with ceremonies and funerals, and ensure that employees are adequately supervised, and proper conduct is maintained. In the absence of the Superintendent, the Contractor must appoint an English-speaking crew foreman or an employee who must be responsible to ensure that the work is being accomplished in an expeditious manner, is performed in accordance with the contract specifications and that the work must progress without undue delay.
- B. Work Hours: Work shall be performed between the hours of 8:00 a.m. to 4:30 p.m., Monday through Friday, except for Holidays, unless otherwise directed by the COR. At Contractor request and with the written permission of the COR, work shall also be permitted to be scheduled for weekends and/or Holidays. When emergency situations, that are caused by the Contractor or severe adverse weather prohibits work during the week, then he/she must obtain approval from the COR to work on weekends in order to meet the period of performance. The Government will not compensate the Contractor for any alternate work schedules needed for the Contractor to complete all contract work within the specified project duration time frame. No work is permitted on Veterans Day and Memorial Day and the two days immediately prior to Memorial Day for preparation. Notwithstanding, if any work under this contract shall be performed outside of the VA's normal working hours (8:00 a.m. to 4:30 p.m. Monday through Friday), then Contractor must notify the COR at least 48 hours in advance.
- C. Federal Holidays: Memorial Day, Juneteenth, Independence Day, Labor Day, Veterans Day, Thanksgiving Day, Christmas Day, New Year's Day, Martin Luther King, Jr., Washington's Day\*

Federal law (5 U.S.C. 6103) establishes the following public holidays for Federal employees. Please note that most Federal employees work on a Monday through Friday schedule. For these employees, when a holiday falls on a non-workday -- Saturday or Sunday -- the holiday usually is observed on Monday (if the holiday falls on Sunday) or Friday (if the holiday falls on Saturday).

*\*This holiday is designated as "Washington's Birthday" in section 6103(a) of title 5 of the United States Code, which is the law that specifies holidays for Federal employees. Though other institutions such as state and local governments and private businesses may use other names, it is our policy to always refer to holidays by the names designated in the law.*

- D. After Normal Hours On-Call/Emergency Situations: The Contractor must establish and maintain a point-of-contact to receive emergency calls from the COR(s). The point-of-contact must be available on a 24-hour basis including during weekends, Federal Holidays and after normal hours of operation. The Contractor must provide name/telephone number(s) for Project Manager (Home & Office), Superintendent, and Foreman, including normal and emergency/after-hours contact numbers, mobile, and fax numbers. In the event of an emergency situation, repairs must be made as expeditiously as circumstances allow or within (24) hours upon initial emergency call. If work is authorized to be performed after hours or on weekends/holidays and an emergency should occur, the Contractor must contact the COR and the Cemetery Director. In the absence of the COR and the Cemetery Director, the Contractor must contact the local police.

## **1.7 WORK SCHEDULE**

- A. Two weeks after Notice to Proceed, the Contractor must provide a Work Plan in the form of a critical path timeline schedule (i.e., Gantt chart) includes, at a minimum, the date of Notice to Proceed, the submittal period, the mobilization date, the start and completion dates of each major work element, showing sequence of section renovations listing each task and subtask, and total time required to complete the renovation, testing and inspection dates, punch list/correction period, demobilization, project completion date, and subsequent maintenance periods. The schedule should prove that the work will be completed within the allotted time for completion of the contract.
- B. Submit proposed Work Plan in electronic and hard copy formats for COR review and approval prior to start of project. The Work Plan must make considerations for weather delays, planting dates, Federal holidays and burial services. The schedule accounts for all work, is logical in its progression through the Cemetery, and is realistic and feasible given daily operations and delays which occur at the cemetery. The Work Plan must be updated weekly to accurately reflect the work that has been completed and any adjustments to milestones and target completion dates. The updated Work Plan must be distributed to the project team and presented to the COR prior to the beginning of weekly progress meetings.

## **1.8 PROJECT MEETINGS and REPORTING:**

- A. The Contractor must report each workday to the Cemetery Director or COR at the main office of the Fort Snelling National Cemetery. Contractor must log in

and obtain funeral and/or special schedules as defined herein. This check in is mandatory. Daily meetings will be at an agreed upon time between the COR and the Contractor. The Contractor must provide an anticipated work schedule each workday or as agreed upon with the Cemetery Director or COR, and to ensure that no work is being performed at the immediate site of a scheduled interment or ceremony. These daily meetings shall also be an opportunity for the Contractor to ask questions and ensure he/she understands the off-limit areas, which will vary, depending on the event. The Contractor's site manager can thus assign tasks accordingly throughout the rest of the Cemetery, so that productive use of labor and equipment is assured, and downtime is avoided. If the Contractor fails to re-direct employees away from an event in a timely fashion, the COR(s) will then assist in doing so. Daily meetings shall be at an agreed upon time between the COR and the Contractor.

- B. A list of scheduled ceremonies will be provided the week prior to the event, and a list of scheduled funerals will be provided daily. The Contractor must be solely responsible for ensuring that no contract work causes any funeral, ceremony, procession or visitation to be delayed, altered, or otherwise impacted in such a way that the dignity or security of the event is compromised. The Contractor must be solely responsible for staying abreast of all such upcoming events and when in doubt, he/she must ask the COR(s).
- C. The Contractor site manager must provide WEEKLY and DAILY work schedules to the CO and COR. The contractor is required to schedule all required services and is further required to comply with their schedule except for delays beyond his/her control. Such delays shall be coordinated through the CO. The work schedule must include deficient work identified by contractor's quality control inspections and not yet corrected.
- D. The Contractor must submit the WEEKLY schedule every Friday by 3:00 P.M. indicating the work to be performed during the following week and the DAILY schedule must be submitted each workday by 8:00 A.M. indicating work being performed on that day.
- E. The Contractor must meet with the COR or authorized designee, at the end of each workday to present documents showing services performed and ensure that work is on schedule.
- F. The Contractor's site manager must provide weekly an accurate written report identifying all work that took place within the previous seven calendar days. The Contractor must document services performed, indicate the location where work was to be performed, and provide information to the COR as required. The report must include an updated Project Work Plan to prove that the work is progressing according to schedule. (See Article 1.8 – "Phasing and Work Sequencing" for requirements of the Project Work Plan.)

- G. In instances where the COR determines that the work is behind schedule, he/she will notify the Contracting Officer and the Contractor must increase workforce and/or hours of operation at no additional cost to the Government in order to achieve completion of the contract work within the specified timeframe.
- H. The VA will set-up and conduct regular virtual Project Meetings. The frequency of these meetings shall be at least weekly or as determined by the CO. The CO, COR, District Project Manager are expected to attend these meetings. On the Contractor's side, the Contractor's project manager and site manager at a minimum must attend these meetings. The Contractor's subcontractor(s) may be asked to attend as appropriate. Be prepared to discuss matters such as procedures, progress, problems, scheduling, coordination, changes, discrepancies, submissions, etc. The VA will record and distribute meeting minutes to all parties.

## **1.9 SCHEDULE OF VALUES**

- A. Two weeks after Notice to Proceed, the Contractor must provide the CO and COR with a Schedule of Values for each major work element in the contract to assist the COR with verification of work performed and to justify progress payments for work completed.

## **1.10 CLEAN UP and DISPOSAL:**

- A. The Contractor must be responsible for cleaning cemetery structures, headstones, monuments, and roadways that are soiled or stained as a result of Contractor's performance. The Contractor must wash-down with water all soiled or stained structures headstones, and monuments at the end of each workday. Roadways must be cleaned with a street sweeper each day as needed to keep the existing roads free from dirt and mud resulting from Contractor operations. No hazardous chemicals shall be used at any time on Government property. The Contractor must bear all costs associated with washing and cleaning. Any such washing/cleaning must be brought to the immediate attention of the COR prior to washing/cleaning.
- B. The Government will not provide receptacle(s) for disposal of debris related to this contract. The Contractor shall be permitted to place his/her trash receptacle dumpsters in the COR approved "Contractor Staging Area".
- C. At the end of each day the Contractor must remove all debris from the cemetery site resulting from the work. The Contractor must ensure at all times that rubbish and trash generated by the Contractor is kept clear of vehicular and pedestrian traffic throughout the site.
- D. The Contractor must dispose of all hazardous waste in accordance with the Resource Conservation and Recovery Act (RCRA) and all other applicable

Federal, State, and Local law and regulations. The Contractor is required to obtain required disposal permits from the State regulatory agency. On-installation storage or disposal of hazardous waste is NOT allowed.

#### **1.11 WORK PERFORMED BY OTHERS ON SITE:**

- A. The Government may undertake or award other contracts for additional work at or near the site of work for this contract. The Contractor must fully cooperate with any other contractors and with Government employees and must carefully adapt scheduling and performance of work, and needing any direction, it will be provided by the COR(s). The Contractor must be responsible for reporting to the COR(s) any problems or questions that shall arise with any other personnel on site during the period of performance of this contract. The Contractor must not take it upon him or herself to resolve any problems or issues with other on-site contractors or employees, but rather must leave it to the COR(s) to resolve the issue.
- B. The Contractor (including his/her employees, subcontractors, consultants or the like must not operate trucks, tractors, and other heavy equipment on any turf area except as provided in this contract to perform work or as authorized by the COR. The Contractor must be responsible for repairing turf damage caused by the Contractor's equipment and staff at no additional cost to the Government.

#### **1.12 WORK ENVIRONMENT:**

- A. All work under this contract shall be performed primarily outdoors and personnel performing these services shall be exposed to wind, sun, cold, dampness, frost, fog and rain. These conditions, may, at times, be extreme. The contractor must take all necessary precautions to protect his/her employees from the elements to the maximum practicable extent. Weather shall not be considered an excusable delay in meeting specifications or project schedule. The Contractor understands that it shall be necessary to work throughout all weather conditions (unless highly extreme, causing equipment rutting and/or hazards, damage to turf, and dangerous to employees or public) and to apply additional labor and equipment as necessary to meet deadlines, at no additional cost to the Government. If weather conditions are so extreme such that work performed on that day may cause more damage to the cemetery grounds than good, (e.g., equipment rutting) and/or posing hazards to employees or public, the COR has the authority to stop work until conditions improve and the COR directs continuation of performance.
- B. Due to the sensitive mission of the cemetery, the work could occasionally involve contact with and/or exposure to grieving individuals. Contractor personnel must exercise and exhibit absolute decorum, composure and stability at all times and refer such individuals to Cemetery Staff.

#### **1.13 CONTRACTOR-FURNISHED ITEMS:**

- A. The Contractor must be responsible for supplying all equipment, personnel, tools, supplies and materials necessary to perform the services required by this contract.
- B. Contractor-furnished items necessary to perform work as required under this contract must be furnished, maintained and operated by the contractor and must be consistent and fully compliant with all applicable OSHA, Federal, State, County, City laws, ordinances and regulations.
- C. The Contractor must be responsible for the supply, maintenance and repair of all contractor-owned equipment. This includes minor maintenance/repair and minor operating parts for equipment such as lubrication, oil changes, spark plugs, gaskets, cotter pins/keys, electric extension cords, etc., to keep all equipment in good operational condition throughout the period of performance of this contract. No contractor equipment maintenance repairs of any kind can be done on cemetery property without COR(s) approval.
- D. Contractor Staging Areas: Primary staging area shall be in the Cemetery spoils area, or as directed by the COR. The Contractor must install a 6-foot high chain link fence with privacy screen fabric and gates around this entire area for duration of the work in those sections. Contractor must base operations out of these staging areas, using it for materials, equipment storage, administration, employee toilets, trash dumpster area, employee parking, employee lunch/break area, etc. After completion of project work, Contractor must remove fencing and restore the area back to original condition.
- E. The Contractor must be responsible for ensuring that all of his/her motor vehicles and equipment meet State inspection, safety, licensing, registration, and insurance requirements.
- F. Only Contractor equipment and supplies (property) required to perform work under this contract must be stored at the Contractor's designated storage area at the end of each workday. No personal property, equipment, or vehicles shall be stored on the cemetery premises.
- G. Electricity: In case the Contractor requires electricity while in performance of this contract, the Contractor shall provide and utilize portable generators as necessary to complete the work. If Contractor hookup to Fort Snelling National Cemetery electrical distribution system is approved by COR, the Contractor shall provide all necessary connections and complete connection work in accordance with NEC requirements, and at no additional cost to the Government.
- H. Water: Water will only be made available via Contractor connection to existing Cemetery water system in existing locations where connection is available, at no cost to the Contractor for purposes of this contract only. The Contractor shall be

responsible for making connections to this water supply in accordance with local regulations. Coordination and COR-approval is required for water connection assemblies. Water outlets must be equipped with shut-off valves to prevent water waste. Contractor must not obstruct normal traffic flow on adjacent roads while water is being obtained. In locations where Cemetery water system is not readily available, the Contractor must provide and utilize portable water tanks and/or water trucks as necessary to transport water to areas where needed in order to complete the work required by this contract, at no additional cost to the Government.

- I. Refuse Facilities: The Contractor must provide his/her own REFUSE FACILITIES, and these shall only be located in the COR approved "Staging Area". The Contractor shall not be authorized to place dumpsters or refuse facilities in any other location at the cemetery other than the approved staging area. The contractor shall be required to dispose of all debris and other waste materials generated by his/her work at a licensed off-site landfill unless otherwise directed by the COR. The Government will not provide receptacles for disposal of debris as a result of the services provided under this contract. In case of Hazardous Materials contractor is required to obtain required disposal permits from the State regulatory agency.
  - J. Portable Toilets: The Contractor must provide portable chemical toilets for use by workmen as necessary to comply with applicable OSHA requirements. These portable toilet facilities must be kept clean and free of excessive odors, insects, etc. Locate portable toilet facilities in the COR approved Contractor Staging Area only. It shall not be permissible to locate portable toilet facilities in any other locations throughout the cemetery. The use of the permanent cemetery toilet facilities by workmen is strictly prohibited throughout the duration of this contract. All connections and appliances connected therewith must be removed prior to completion of contract, and premises left perfectly clean. The Contractor and Contractor's workers must not use Cemetery restroom facilities.
  - K. Identification, Parking, Smoking, and VA Regulations: The Contractor's employees must wear visible identification at all times while on the premises of the Cemetery. It shall be the responsibility of the Contractor to park in the appropriate designated parking areas. The Cemetery will not invalidate or make reimbursement for parking violations of the Contractor under any conditions. Smoking shall be prohibited inside any buildings at the Cemetery. Possession of weapons is prohibited. Enclosed containers, including tool kits, shall be subject to search. Violations of VA regulations will result in citation answerable in the United States (Federal) District Court, not a local district, state, or municipal court.
1. The Contractor must be responsible to ensure Contractor employees providing work on this contract must be fully trained and completely competent to perform the required work.

#### **1.14 STANDARDS OF CONTRACTOR AND EMPLOYEE CONDUCT:**

- A. Contractor personnel must be required to adhere to the following standards of dress, conduct, supervision and training while performing work on Government property. These standards and regulations are enforceable under Title 38, U.S.C., Part I, Chapter 9, Section 5901. It will be subject to immediate enforcement action by the Contracting Officer if these standards are not adhered to during the period of performance of this contract. Contractor management must be responsible for training and safety precautions prescribed by OSHA regarding safety equipment and devices.
- B. Contractor personnel must be fully clothed at all times, to include upper garment to cover body from the waist to the neck and long pants or slacks. Clothing must be clean, and presentable and suitable to the work while maintaining proper appearance and decorum indicative for a National Shrine. Overall cleanliness and personal hygiene are imperative. Uniform shirts and hats are preferred. If uniforms are used, they must be in unison among all employees. Garments, which have a message, slogan or printing of any kind other than the Contractor's business attire, shall be prohibited. T-shirts and/or tank tops as outer garments are prohibited. Protective/safety clothing and shoes must meet or exceed OSHA and state requirements. The Contractor must maintain a neat and professional appearance throughout its workforce, vehicles, equipment, and maintenance areas.
- C. The Contractor/personnel must not engage in loud or boisterous behavior, angry outbursts or use profane or abusive language at any time on Government premises. Playing radios and/or electronic games/devices is prohibited. Due to the sensitive mission of the cemetery, Contractor employees shall come into daily contact with grieving individuals, therefore Contractor employees must exercise and exhibit absolute decorum, courtesy, and respect while within the cemetery or at its perimeter or entrances. Inquiries from cemetery visitors must be politely referred to Government cemetery staff. Gratuities of any kind must be strictly prohibited.
- D. Contractor personnel must consume food and beverage only within areas designated by the COR(s). Intoxication, and violence or criminal acts of any kind will not be tolerated, and is cause for immediate removal from Government property. Use or sale of intoxicating beverages and/or drugs is strictly prohibited (including legalized marijuana use). Use of tobacco products is only allowed in specific areas designated by the COR/Cemetery Director.
- E. Contractor employees must only take breaks/rest periods and lunch breaks at the Contractor Staging Area, not in the field. Eating and drinking (except water) is prohibited in the work areas and within sight of a committal shelter during a service.

- F. Workers must utilize contractor provided temporary chemical toilet facilities located only in the Contractor Staging Area. Misconduct shall form the basis for immediate contract enforcement action, to include immediate removal from the cemetery.
- G. No tools, equipment or other items will be placed or leaned on headstones or monuments. Contractor personnel must not lean, sit, or stand on or against headstones or monuments. Nor can they drive over them. Contractor personnel must use tools approved by the Contracting Officer Representative (COR), to lift markers out of the ground; pickaxes are not an acceptable tool for moving markers.
- H. Identification, Parking, Smoking, and VA Regulations: The Contractor's employees must wear visible identification at all times while on the premises of the Cemetery. Contractor employees may park privately owned vehicles in the area designated for parking by the COR. It is the responsibility of the Contractor to park in the appropriate designated parking areas. The Cemetery will not invalidate or make reimbursement for parking violations of the Contractor under any conditions. Workers are not allowed to park throughout the cemetery grounds, and violation of this requirement shall form the basis for immediate contract enforcement action, to include immediate removal from the cemetery. Smoking is prohibited inside any buildings at the Cemetery. Possession of weapons is prohibited. Enclosed containers, including tool kits, shall be subject to search. Violations of VA regulations may result in citation answerable in the United States (Federal) District Court, not a local district, state, or municipal court.
- I. Orientation for Contractor Employees:
  - 1. The Contractor will be responsible to ensure that Contractor employees coming to the work site will receive complete information on each of these subjects:
    - a. Safety and Environmental Protection, and Fire Safety.
    - b. Project Work Schedule.
    - c. Rules Pertaining to Workers.
    - d. General Parameter Job Related Issues.
    - e. Disaster procedures.
    - f. All technical requirements of the project.
  - 2. The Contractor will be responsible to ensure Contractor employees providing work on this contract are fully trained and completely competent to perform the required work.
  - 3. The Contractor must be responsible to ensure that his/her employees (including contractor's consultants, subcontractors, etc.) are aware of all the terms and conditions set forth above in this solicitation package regarding

their performance and conduct during the performance period of this contract. Each contractor and subcontractor employee must sign a Statement of Compliance, (Attachment 8); the Contractor must deliver the signed statements to the COR before work may begin or before any new employee is allowed to work on Cemetery grounds

**J. Labor Force and Equipment:**

1. The Contractor must be aware of the intensive labor and equipment requirements needed to meet contract specifications. Contractor must be responsible to provide, labor and equipment as necessary to meet deadlines at no additional cost to the Government. The Contractor must provide additional resources necessary to meet or return to specified work requirements after special events and/or services. The Contractor must determine the number of employees and the amount and kinds of equipment needed during the period of performance.
2. Employee Listing: The Contractor must maintain and provide the COR a current list of all employees on site including subcontractor personnel. The list must include the employee's name, job title, and driver's license number.
3. Employee Identification: The Contractor's employees must wear visible identification at all times while on the premises of the Cemetery.

**1.15 SAFETY and ENVIRONMENTAL PROTECTION:**

- A. The Contractor is responsible to ensure that all work must be done in a manner that safeguards all VA visitors, employees, and public. The Contractor must be solely responsible for any and all actions initiated and/or completed by his/her employees. Furthermore, the contractor and his/her employees must have a clear understanding of, and be sensitive to, such environmental issues as ground water contamination, wetlands, etc., and be consistent and fully compliant with all applicable Federal, State, County and City laws, ordinances, Right-to-Know laws, EPA guidelines, and regulations.
- B. Matters related to safety, and any actions of the Contractor, must meet all safety requirements of Fort Snelling National Cemetery's Safety Officer, Department of Veterans Affairs, OSHA, and the State. It is incumbent upon the Contractor to be familiar with these requirements. "Safety" must also include the Contractor having a safety representative who maintains regular and routine contact with the Safety Officer at Fort Snelling National Cemetery.
- C. Adequate warning devices, barricades (orange construction fence), guards, flagmen or other necessary precautions must be provided by the Contractor at all work sites to eliminate hazards and for the protection, safety, and warning of all public visitors, pedestrians, cemetery employees, and vehicular traffic within the area. All open and unattended holes in the ground must be covered with

plywood and/or barricaded for pedestrian safety. All work areas will be secured with barricades (orange construction fence) at the end of each workday.

- D. Contractor fuel storage containers must be limited to those less than 55-gallon storage capacity. Any fuel storage must comply with environmental laws and regulations.
- E. Spill Prevention, Containment, and Clean-up: The Contractor must contain, clean up, and report all spills on Government property in a manner that complies with all applicable Federal, State, and Local laws and regulations or otherwise stated herein, and the installation spill control plan at no cost to the Government.
- F. Dust Control: Keep dust down at all times, including during nonworking periods. Sprinkle or treat with dust suppressants, the soil at the site, haul roads, and other areas disturbed by operations. Dry power brooming will not be permitted. Instead, use vacuuming, wet mopping, wet sweeping, or wet power brooming.

#### **1.16 FIRE SAFETY:**

- A. Applicable Publications: Publications listed below form part of this Article to the extent referenced. Publications are referenced in text by basic designations only.
  - 1. National Fire Protection Association (NFPA):
    - a. 10-2013 Standard for Portable Fire Extinguishers
    - b. 70-2017 National Electrical Code
  - 2. Occupational Safety and Health Administration (OSHA)
    - a. 29 CFR 1910 Safety and Health Regulations for Personal Protection, Safety Color Codes, Portable Power Tools, Electrical Safety and Portable Electric Equipment.
- B. Fire Safety Plan: Establish and maintain a fire protection program in accordance with 29 CFR 1910. Prior to start of work, prepare a plan detailing service-specific fire safety measures, including periodic status reports and submit to COR for review.
- C. Site and Building Access: Maintain free and unobstructed access to facility emergency services and for fire, police and other emergency response forces in accordance with NFPA 241.
- D. Separate temporary facilities, such as trailers, storage sheds, and dumpsters, from existing buildings and new construction by distances in accordance with NFPA 241. For small facilities with less than 6 m (20 feet) exposing overall length, separate by 3m (10 feet). Locate only in staging area where approved by COR.

- E. Temporary Electrical: Install, use and maintain installations in accordance with 29 CFR 1910, NFPA 241 and NFPA 70. If temporary electrical is required, the Contractor must notify the COR prior to use of Temporary Electrical Services.
- F. Means of Egress: Do not block exiting for occupied buildings, including paths from exits to roads.
- G. Fire Extinguishers: Provide and maintain extinguishers in service areas and temporary storage areas in accordance with 29 CFR 1926, 29 CFR 1910, NFPA 241 and NFPA 10.
- H. Flammable and Combustible Liquids: Store, dispense and use liquids in accordance with 29 CFR 1910, NFPA 241 and NFPA 30.
- I. Smoking: Smoking is prohibited except in designated smoking areas.

#### **1.17 OPERATIONS AND STORAGE AREAS:**

- A. Working space and space available for storing materials is to be where approved by COR in the Contractor's "Staging Area". The Contractor must confine all operations (including storage of materials) on Government premises to areas authorized or approved by the COR or Contracting Officer.
- B. The Contractor must hold and save the Government, its officers and agents, free and harmless from liability of any nature occasioned by the Contractor's performance. It is understood that the VA will not be held responsible for any damage to the Contractor's equipment, materials, supplies or the like that result from vandalism, theft etc. while on site.
- C. Temporary buildings (e.g., storage sheds, shops, offices) and utilities may be erected by the Contractor only with the approval of the COR and shall be built with labor and materials furnished by the Contractor without expense to the Government. The temporary buildings and utilities shall remain the property of the Contractor and must be removed by the Contractor at his/her expense upon completion of the work.
- D. The Contractor must, under regulations prescribed by the Contracting Officer or COR, use only established roadways, or use temporary roadways constructed by the Contractor when and as authorized by the COR or Contracting Officer. Vehicles must not be loaded beyond the loading capacity recommended by the manufacturer of the vehicle or prescribed by any Federal, State, or local law or regulation. When it is necessary to cross curbs or sidewalks, the Contractor must protect them from damage. The Contractor must repair or pay for the repair of any damaged curbs, sidewalks, or roads.

- E. Execute work in such a manner as to interfere as little as possible with work being done by others. Keep roads clear of service materials, debris, standing service equipment and vehicles at all times.
- F. Do not store materials and equipment in other than assigned areas.
- G. Fence: Before work operations begin, Contractor must provide a chain link fence, six feet minimum height, around the staging area. Provide gates as required for access with necessary hardware, including hasps and padlocks. Fasten fence fabric to terminal posts with tension bands and to line posts and top and bottom rails with tie wires spaced at maximum 15 inches. Bottom of fences must extend to one inch above grade. Fence fabric must be covered with HDPE privacy screen. Screen color must be dark green or black with 85% minimum density sufficient to obscure view of the items within. After completion of project work, Contractor must remove fencing and restore area back to original condition.
- H. To minimize interference of service activities with flow of Cemetery traffic, comply with the following:
  - 1. Keep roads, walks and entrances to grounds, to parking and to occupied areas of buildings clear of materials, debris and standing service equipment and vehicles. At least one lane must be open to traffic at all times.
  - 2. Coordination of work with COR/Cemetery Director or authorized designee: The burial activities at a National Cemetery will take precedence over Contractor activities. The Contractor must cooperate and coordinate with the COR/Cemetery Director or authorized designee, in arranging schedule to cause the least possible interference with cemetery activities in actual burial areas. Work noise during the interment services must not disturb the service. Trucks and workmen must not pass through the service area during this period.

#### **1.18 THE GOVERNMENTS RESPONSIBILITIES:**

- A. Upon award of the contract, the Government will inform the Contractor prior to commencing the work, of any known damages to the cemetery grounds, headstones/markers or any other areas that the Contractor is unaware of and not responsible for. In addition, upon award of the contract, a walk-through of the cemetery grounds will be scheduled by mutual agreement of the Government and Contractor, and will be attended by the COR, and appropriate contractor personnel (to include the Contractor Supervisor and Project Manager). The Contracting Officer is an optional attendee.
- B. The Government will not provide the Contractor with any furnishings, fuel storage, equipment, materials, restrooms, or telephones.

- C. Notification of Non-Compliance: The COR will notify the Contractor of any detected non-compliance with the foregoing requirements. The Contractor must take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, shall be deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order ceasing all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such cessation direction shall be made the subject of claim for extension of time for excess costs or damages by the Contractor.

#### **1.19 PHASING AND WORK SEQUENCING:**

- A. Contract work must be accomplished in a sequenced manner, with work limited to occurring in no more than one contiguous area at any given time in order to minimize overall disruption to the cemetery. Requests to work on more contiguous area at a time may be submitted to the COR for approval. Approval/disapproval will be based on the impact to the cemetery operations and the progress and workmanship of the contractor, as determined by the COR. Work should not proceed into subsequent areas until such time that that work has been completed in prior areas. Submit proposed project Work Plan sequence for COR review and approval prior to start of project.
- B. The effective period of construction shall occur between March and November of each year weather permitting. Turf must be established prior to the onset of winter season. Bare ground is not acceptable during this time.

#### **1.20 USE OF CEMETERY FACILITIES:**

- A. The Government shall not be responsible for any loss, damage, or theft of Contractor items. Contractor shall be responsible for acceptable standards of housekeeping and custodial maintenance of cemetery facilities used by Contractor's employees.
- B. The Government will not furnish a storage building at the Cemetery site for use by the Contractor to store supplies and equipment. However, the Government will provide an area designated for the Contractor's use. The Contractor can establish facilities to include but not limited to office site, covered storage, portable toilet facilities etc. on or in the designated area after such facilities are approved by the COR and Contracting Officer (CO). All utilities to this area are the responsibility of the Contractor. The Government will not be responsible for any damage to or loss of the Contractor's equipment and supplies stored on the Government's premises. The Contractor shall be responsible for maintaining fire extinguishers and other safety equipment.

- C. The Contractor will be responsible for safely storing any chemicals, pesticides, herbicides, cleaning solutions, etc. in accordance with manufacturers recommendations. An SDS (Safety Data Sheet) is required for all chemicals, pesticides, herbicides and cleaning solutions.
- D. Electricity and phone service will NOT be furnished by the Government for the Contractor's work area. (See "Contractor-Furnished Items".)

#### **1.21 SUPERVISION AND TRAINING:**

- A. The Contractor must provide a supervisor who speaks and writes fluent English on site, and at all times when Contractor personnel are on the premises.
- B. The Contractor must be responsible for maintaining satisfactory standards of personnel conduct and work performance and must administer disciplinary action as required. The Contractor is expected to remove any employees from the Cemetery for cause, to include, but not limited to, safety violations, other misconduct in performance of duty under these specifications and/or conduct contrary to the best interests of the Government. If the Contractor fails to act in this regard, or the reason for a removal is immediately required to protect the interests of the Government, the COR may direct the removal of an employee from the premises. Contractor objections to any such action will be referred to the Contracting Officer (CO) for final resolution; however, the Contractor will first immediately comply with COR direction pending any CO final resolution at a later time or date. The Contractor will not be due any type of compensation for their costs incurred as a result of an employee being removed for cause; unless the removal is directed by the COR, and is later found invalid and/or unreasonable by the Contracting Officer.
- C. The Contractor must also be responsible for training and safety precautions for Contractor employees performing work under these specifications. OSHA standards must be observed by the Contractor in all work performed. Contractor must ensure that appropriate safety equipment is used by Contractor personnel and must be used as prescribed by OSHA standards, including hard hats, safety shoes, safety glasses, and hearing protection devices. The following OSHA and NFPA standards and codes are to be adhered to:
  - 1. National Fire Protection Association (NFPA): 10-2013 Standard for Portable Fire Extinguishers
  - 2. Occupational Safety and Health Administration (OSHA) 29 CFR 1910 Safety and Health Regulations for Personal Protection, Safety Color Codes, Portable Power Tools, Electrical Safety and Portable Electric Equipment.

3. Site and Building Access: Maintain free and unobstructed access to facility emergency services and for fire, police and other emergency response forces in accordance with NFPA 241.

**1.22 INSPECTION AND CLEANING OF CEMETERY FACILITIES:**

- A. The Contractor will perform a weekly inspection. During this inspection the appearance of the Cemetery will be observed, and any deficiencies from the contract will be noted. Deficiencies must be corrected as soon as practicable. Items that need correcting outside the scope of the contract will be reported to the COR or their representative.
- B. The Contractor will be required to submit inspection reports and work accomplished to the COR weekly. The inspection forms will be provided to the Contractor. (See Attachments 1 and 2 – “Work Summary and Progress Report”).

**1.23 INSTALLATION DURING CEMETERY FUNCTIONS:**

- A. Contractor personnel must not operate motorized equipment or conduct other commercial activities within the designated area during interment services. The COR will identify the designated area. The COR or their representative shall furnish the Contractor with a schedule of all interments and/or ceremonies no later than the close of business of the day prior to the scheduled interment, and a minimum of three (3) days before any ceremonial events.

**1.24 PERFORMANCE EVALUATION MEETING:**

- A. The issuance of a Contract Discrepancy Report (CDR), found at Attachment 4, may be cause for the scheduling of a meeting among the Contractor, Contracting Officer, and the COR. A mutual effort will be made to resolve all problems identified. The Government will prepare written minutes of the meeting. The Contractor, Contracting Officer, and the COR will sign minutes of the meeting(s).
- B. Should the Contractor not concur with the minutes, they will so state their objections in writing to the Contracting Officer, within ten calendar days, and also explain the reasons for non-concurrence. The Contracting Officer will review and consider the reasons submitted for the Contractor's non-concurrence and make a decision to revise or uphold the comments as written. The Contracting Officer will notify the Contractor of the decision in writing within ten calendar days of submission of any objections.

**1.25 CONTRACTOR RESPONSE TIMES:**

- A. During Normal Working Hours: The Contractor must be prepared to shift workers and work areas as necessary to safeguard the cemetery and its visitors and restore the work area(s) to a safe condition. The Contractor must respond to all normal working hours' requests made by the COR(s) within 30 minutes of the initial notification.

- B. After Normal Hours: After Hours requests warrant rapid response in order to safeguard the cemetery and its visitors and restore the work area(s) to a safe condition. The Contractor must respond to all after normal hours requests made by the COR(s) within one (1) hour of the initial call.
- C. Calls by other Government employees other than the COR(s) shall not be acted upon, but shall be referred to the COR(s) for review. Repairs must begin as expeditiously as circumstances allow and should normally be completed within (24) hours of the initial emergency call. The Contractor must keep the COR(s) fully informed of the situation and what action will be taken to secure and correct the situation.

**1.26 CONTRACTOR'S QUALITY CONTROL PLAN (QCP):**

- A. The Contractor must establish and maintain a complete QCP to assure the requirements of this contract are provided as specified. This QCP will be forwarded to the Contracting Officer along with the requested initial proposal and may be revised as necessary during the life of the contract, with concurrence of the COR and Contracting Officer. The Contracting Officer will review the QCP and list any needed clarifications, and return to Contractor for response, if necessary. The Contractor's QCP must include the following or have incorporated into during performance of contract, at a minimum:
  - 1. An inspection plan covering all services required by this contract. The inspection plan must specify the areas to be inspected on either a scheduled or unscheduled basis, how often inspections will be accomplished and documented, and the title of the individual(s) who will perform the inspections.
  - 2. On-site records of all inspections conducted by the Contractor noting necessary corrective action taken. The Government reserves the right to request copies of any and/or each inspection.
  - 3. Incorporation of either active or established internal policy and procedures for updating equipment and procedures that may affect performance of contract.
  - 4. The methods for identifying and preventing deficiencies in the quality of service performed, before the level of performance becomes unacceptable; and the organizational functions, intermediate supervisory responsibilities, and overall management responsibilities for ensuring total acceptable performance.
  - 5. On-site records identifying the character, physical capabilities, certifications and ongoing training of each employee performing services under this contract.

6. A log to account for all requests for immediate service. The log must indicate the date and time of services, and description of results and completion of these services.
7. On-site records of any complaints or problems, with procedures taken to allow for corrections and/or elimination before effects caused interruption of performance of contract.

**1.27 QUALITY ASSURANCE:**

- A. The COR will evaluate the Contractor's performance through on-site inspections, evaluation of the Contractor's quality control program and receipt of complaints from cemetery personnel.
- B. The COR will also receive and investigate complaints from various customers visiting the Cemetery. The Contractor must be responsible for initialing validated visitor complaints. The COR shall make final determination of the validity of visitor complaint(s).
- C. The COR will initially inspect each task in the process of Cemetery Section Renovation as work progresses utilizing Attachment 3 – "Contract Performance & Quality Assurance Report". The COR will note in the report if the work "did or did not" meet the requirements of the contract and corrective actions needed. The COR will provide a copy of the report to the CO and Contractor for further action.
- D. The COR may inspect each task as completed or increase the number of quality control inspections if called for by repeated failures discovered during inspections or repeated customer complaints. Likewise, the COR may decrease the number of quality control inspections if performance dictates.

**1.28 ACTIONS:**

- A. Normally, the COR will verbally advise or give a written inspection report to the Contractor of discrepancies the first time they occur and ask the Contractor to correct the problem. A notation will be made on the COR checklist of the date and the time the deficiency was discovered and the date and time the Contractor was notified.
- B. If the Government created any of the discrepancies, these will not be counted against the Contractor's performance. When the Government has caused the Contractor to perform in an unsatisfactory manner, the COR will forward a written notice to the responsible organizational element requesting corrective action be taken.
- C. When the Contractor is not meeting the acceptable limits of satisfactory performance, a CDR will be issued to the Contractor. The seriousness of the failures should govern whether to issue CDR at the end of the period, or as soon as the limits of satisfactory performance are exceeded.

- D. When a CDR is issued for a service, the Contracting Officer and/or the COR may exercise any contractual remedy available for non-performance, in accordance with FAR 52.212-4, "Inspection and Acceptance".
- E. If the Contractor does not achieve satisfactory performance by the end of the next period or agreed suspense date, further actions may be considered, to include a determination on whether continued performance by the contractor is feasible.

**1.29 HISTORIC PRESERVATION:**

- A. Where the Contractor or any of the Contractor's employees, prior to, or during the services work, are advised of or discover any possible archeological, historical and/or cultural resources, the Contractor must immediately notify the COR verbally, and then with a written follow up.
- B. The COR/Cemetery Director may direct the Contractor to follow behind other projects as they are completed, so as to not disrupt work sequence of existing contracts under construction.

**1.30 EXPOSURE OR ACCIDENTAL DISCOVERY OF REMAINS:**

- A. Should any on-site activity, incident, emergency, or disaster result in exposure and/or damage to any remains, container for remains (i.e., casket or urn), or outer burial container, the Contractor must IMMEDIATELY notify the Cemetery Director, the COR and/or the CO for guidance; and stop any further work and await permission to proceed.

**1.31 INSPECTIONS AND SURVEYS:**

- A. PROFESSIONAL SURVEYING SERVICES - A registered professional land surveyor or registered civil engineer whose services are retained and paid for by the Contractor must be used for layout control of work and to restore any grave section corner monuments that may be disturbed as a result of the Contractors performance of the contract work.
- B. Survey: Before any work is started, the Contractor must make a thorough digital video survey with the COR of areas in which contract work occurs and areas of anticipated routes of access, etc., and provide a copy to the COR. This digital video survey must include:
  - 1. Existing conditions at site. Use a video camera to survey and document existing conditions prior to start of work. Contractor is to provide the digital video camera and memory media for this purpose. The Contractor must turnover one copy of the completed video survey data to COR prior to starting work. The video survey must sufficiently document the pre-construction alignment of headstones in adjacent areas to use as a comparison to the post-construction alignment documentation.

2. Re-Survey: At the final inspection and acceptance the Contractor and COR together shall make a thorough re-survey of the areas where contract work occurred. Re-survey report must list any damages caused by Contractor's workmen in executing work of this contract.
- C. Daily Log of Construction report: Contractor must prepare a daily construction log and submit them weekly to the CO, COR and District PM for all construction related work, including representative photos documenting work accomplished. (i.e., submit 5 separate daily reports once a week.) Contractor may use a standardized daily construction log form acceptable to the COR, or the VA-provided form. (See Attachment 9 Contractor's Daily Log of Construction Form.)

**1.32 PROTECTION OF EXISTING VEGETATION, STRUCTURES, EQUIPMENT, UTILITIES, AND IMPROVEMENTS:**

- A. Damage to Government property: The Contractor must be responsible for repair or replacement of any contractor damaged cemetery structure, to include: turf, curb, road pavement, headstones or markers, irrigation equipment, valve boxes, grid monument control markers, trees, plant beds, etc., which are chipped, marred, damaged and/or ruined at the fault of the Contractor and must bear all costs associated with replacement and reinstallation. Any such damage must be brought to the immediate attention of the appointed Contracting Officer Representative (COR) prior to repair/replacement/installation.
- B. The Contractor shall preserve and protect all structures, equipment, and vegetation (such as trees, shrubs, and grass) on or adjacent to the work site, which shall not be removed, and which does not unreasonably interfere with the work required under this contract. The Contractor shall only remove trees when specifically authorized to do so by the COR, and must avoid damaging vegetation that will remain in place. If the Contractor is obstructed by tree roots within the drip-line of a standing tree that are 2 inches in diameter or greater, the Contractor must request an inspection by the COR. The COR will inspect and determine if any roots will be pruned. If any limbs or branches of trees are broken during contract performance, or by the careless operation of equipment, or by workmen, the Contractor must trim those limbs or branches with a clean cut and remove and properly dispose of from the site as defined herein.
- C. The Contractor must protect from damage all existing improvements and utilities at or near the work site and on adjacent property of a third party, the locations of which are made known to or are known by the Contractor. The Contractor must immediately notify the COR of any such occurrence and repair any damage to those facilities, including those that are the property of a third party, resulting from failure to comply with the requirements of this contract or failure to exercise reasonable care in performing the work. If the Contractor fails or refuses to repair the damage promptly, the Contracting Officer will have the necessary work performed and charge the cost to the Contractor.

D. Where work conflicts with existing utility/service lines (above ground/below ground), the corresponding utilities company and the COR will be notified and the Contractor must obtain any necessary permits/ blue prints and cooperate with the utilities company/cemetery staff to avoid any damage or liability, and provide a safe work environment for his/her employees. Contractor must be responsible for damages to utilities, above and below ground.

E. Maintenance of Existing Turf:

1. The Contractor must not interfere with maintenance of turf areas not “currently” being renovated. Once the Contractor has moved into a cemetery section, the Contractor must be responsible for maintaining healthy turf where turf renovation is not a part of the contracted work for that section to include: watering, fertilizing, pest control and mowing/edging. Any turf that is allowed to die or become stressed as a result of improper maintenance must be restored by the Contractor at no additional cost to the Government to the satisfaction of the COR. If turf restoration is required due to contractor negligence, the turf must be maintained for an additional period not less than 30-days after being restored to include watering, fertilizing, pest control and mowing/edging. Once section renovation has been accepted by the Government, the turf shall be returned to the care of the Government and/or the Government’s grounds maintenance contractor.

F. Protection of Existing Utilities: Contractor must be responsible for damages to utilities, above and below ground. If the Contractor fails or refuses to repair the damage promptly, the Contracting Officer may have the necessary work performed and charge the cost to the Contractor.

G. Interruption of Utilities Services:

1. No utility service such as water, gas, steam, sewers or electricity, or fire protection systems and communications systems shall be interrupted without prior approval of COR.
2. The Contractor must not interrupt irrigation to areas not “currently” being renovated. If necessary, the contractor must cap off, run by-pass lines or manually irrigate as necessary, as to not affect areas not currently being renovated.
3. Contractor must submit a request to interrupt any such services to COR, in writing, a minimum of 48 hours in advance of proposed interruption. Request must include the reason, date, exact time of, and approximate duration of such interruption.

4. Contractor will be advised (in writing) of approval of request, or of which other date and/or time such interruption will cause least inconvenience to operations of Cemetery. Interruption time approved by COR may occur at other than Contractor's normal working hours.
- H. Refer to Articles, "Inspections and Surveys", "Restoration", and "Operations and Storage Areas" for additional instructions concerning repair of damage to structures and site improvements.

**1.33 RESTORATION:**

- A. Contractor must remove, cut, alter, reinstall, replace, patch and repair existing work as necessary. Except as otherwise shown or specified, do not disturb any water, steam, gas, or electric services without prior approval of the COR or Contracting Officer. Existing work to be completed and that is found to be defective in any way, must be reported to the COR before it is disturbed. Materials and workmanship used in restoring work must conform in type and quality to that of original service, except as otherwise shown or specified.
- B. Upon completion of contract, deliver work complete and undamaged. Existing cemetery features (lawns, paving, roads, walks, etc.) disturbed or removed as a result of performing required new work, must be patched, repaired, reinstalled, or replaced with new work, and refinished and left in as good condition as existed before commencing work. All restoration work must be accomplished without undue delay.
- C. At Contractor's own expense, Contractor must immediately restore to service and repair any damage caused by Contractor's workmen/subcontractors to existing cemetery infrastructure.
- D. Expense of repairs to such utilities and systems not shown on drawings or locations of which are unknown will be covered by adjustment to contract time and price.
- E. The Contractor must be responsible for replacement of any headstones or markers, which are chipped, marred, or damaged at the fault of the Contractor and must bear all costs associated with replacement and reinstallation. Any such damage must be brought to the attention of the COR prior to replacement or reinstallation.

**1.34 LAYOUT OF WORK:**

- A. The Contractor must lay out the work and must be responsible for all measurements in connection with the layout. Where burial section grid monumentation exists, it may be utilized for all headstone row layout work, and contractor must coordinate all associated field layout dimensions with COR prior to start of work. The Contractor must furnish, at Contractor's own expense, all stakes, templates, platforms, equipment, tools, materials, and labor required to

layout any part of the work. The Contractor must be responsible for executing the work to the lines and grades needed to accomplish the work and to ensure that grave markers are correctly and accurately located on their associated gravesites.

- B. The Contractor must also be responsible for maintaining and preserving all temporary and permanent stakes and other marks until authorized by the COR to remove them. If such marks are destroyed by the Contractor or through Contractor's negligence before their removal is authorized, the Contracting Officer will order the contractor to replace them, or deduct the expense of the replacement from any amounts due or to become due to the Contractor.
- C. PROFESSIONAL SURVEYING SERVICES - A registered professional land surveyor or registered civil engineer whose services are retained and paid for by the Contractor must be used for layout control of work and to restore any grave section corner monuments that may be disturbed as a result of the Contractor's performance of the contract work.

**1.35 AS-BUILT DRAWINGS:**

- A. The Contractor must maintain one set of as-built drawings, which must be kept current during performance of the project, to include all contract changes, modifications, and indicating work progress on a section-by-section basis.
- B. Contractor must deliver one set of as-built drawings to the COR within 15 calendar days after acceptance of the project by the Contracting Officer.
- C. Provide as-built drawings of any modifications to the irrigation system. Graphically show pipe routing and dimension pipe locations from two permanent fixed points on the site.

**1.36 DISPOSAL OF HAZARDOUS WASTE:**

- A. The Contractor must not dispose of any hazardous materials at any location on the cemetery premises except as specified and/or authorized by the COR/Cemetery Director.

**1.37 METRIC CONVERSIONS:**

- A. Any measurements stated in this specification or in any documents associated with the proposed contract as normally used inch-pound units can be converted to metric units providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376. Any measurements only given in inch-pound units are not meant to be restrictive. If the General Requirements require replacement of parts to be included as part of the contract, and a product (part) is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request must be made to the Contracting Officer to determine if the product is acceptable.

**1.38 GENDER:**

- A. For the purpose of equal rights, wherever the masculine gender is used in this solicitation, and the resulting contract, it shall be considered to include both masculine and feminine gender.

**1.39 SALES OF COMPANIES OR NAME CHANGES:**

- A. The Contractor, gaining award of the contract, will notify the Contracting Officer (Attn: Department of Veterans Affairs, National Cemetery Administration, Centralized Contracting Division, (41D3A), 74 Barrett Heights Road, Suite 309, Stafford, VA 22556., in writing immediately upon entering an agreement (either oral or written) to sale or transfer all or part of its stock or other ownership interest to any other party. This Contractor must also, contact the Contracting Officer immediately in writing (at the above referenced address) in the event a company name change shall happen, with information as to the reasons and/or determinations causing this action.

**B. SPECIFICATIONS**

|                |                                            |
|----------------|--------------------------------------------|
| <b>SECTION</b> | <b>DIVISION 01 – GENERAL REQUIREMENTS.</b> |
| 01 33 23       | SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES   |
| 01 45 29       | TESTING LABORATORY SERVICES                |
|                |                                            |
| <b>SECTION</b> | <b>DIVISION 02 – EXISTING CONDITIONS</b>   |
| 02 21 13       | SITE SURVEYING (LIMITED)                   |
|                |                                            |
| <b>SECTION</b> | <b>DIVISION 31 – EARTHWORK</b>             |
| 31 00 00       | EARTHWORK                                  |
|                |                                            |
| <b>SECTION</b> | <b>DIVISION 32 – EXTERIOR IMPROVEMENTS</b> |
| 32 39 20       | HEADSTONE SETTING, ALIGNMENT AND CLEANING  |
| 32 92 24       | TURF RENOVATION                            |

**SECTION 01 33 23 - SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES**

**PART 1 - GENERAL**

- 1.1 Refer to Articles titled SPECIFICATIONS AND DRAWINGS FOR

CONSTRUCTION (FAR 52.236-21) and, SPECIAL NOTES (VAAR 852.236-91), in Section, GENERAL CONDITIONS.

- 1.2 For the purposes of this contract, samples (including laboratory samples to be tested), test reports, certificates, and manufacturers' literature and data shall also be subject to the previously referenced requirements. The following text refers to all items collectively as SUBMITTALS.
- 1.3 Submit for approval, all of the items specifically mentioned under the separate sections of the specification, with information sufficient to evidence full compliance with contract requirements. Materials, fabricated articles and the like to be installed in permanent work must equal those of approved submittals. After an item has been approved, no change in brand or make will be permitted unless:
  - A. Satisfactory written evidence is presented to, and approved by Contracting Officer, that manufacturer cannot make scheduled delivery of approved item or;
  - B. Item delivered has been rejected and substitution of a suitable item is an urgent necessity or;
  - C. Other conditions become apparent which indicates approval of such substitute item to be in best interest of the Government.
- 1.4 Submittals must be accompanied by a Submittal Register Log provided by the Contracting Officer.
- 1.5 Forward submittals in sufficient time (a minimum of 10 calendar days, exclusive of mailing time) to permit proper consideration and approval action by Government. Time submission to assure adequate lead time for procurement of contract - required items. Delays attributable to untimely and rejected submittals (including any laboratory samples to be tested) will not serve as a basis for extending contract time for completion.
- 1.6 Submittals will be reviewed for compliance with contract requirements by Contracting Officer, and action thereon will be taken by the Contracting Officer.
- 1.7 Upon receipt of submittals, Contracting Officer will assign a file number thereto. Contractor, in any subsequent correspondence, must refer to this file and identification number to expedite replies relative to previously approved or disapproved submittals.
- 1.8 The Government reserves the right to require additional submittals, whether or not particularly mentioned in this contract. If additional submittals beyond those required by the contract are furnished pursuant to request therefore by Contracting Officer, adjustment in contract price and time will be made in accordance with Articles titled CHANGES (FAR 52.243-4) and CHANGES - SUPPLEMENT (VAAR 852.236-88) of the GENERAL CONDITIONS.
- 1.9 Schedules called for in specifications and shown on shop drawings must be submitted for use and information of Department of Veterans Affairs and District

Engineers. However, the Contractor must assume responsibility for coordinating and verifying schedules. The Contracting Officer assumes no responsibility for checking schedules or layout drawings for exact sizes, exact numbers and detailed positioning of items.

- 1.10 Submittals must be submitted by Contractor only and shipped prepaid. Contracting Officer assumes no responsibility for checking quantities or exact numbers included in such submittals.
- A. Submit samples in single units unless otherwise specified. Submit shop drawings, schedules, manufacturers' literature and data, and certificates in quadruplicate, except where a greater number is specified.
  - B. Submittals will receive consideration only when covered by a transmittal letter signed by Contractor. Letter must be sent via first class mail and must contain the list of items, name of Cemetery, name of Contractor, contract number, applicable specification paragraph numbers, applicable drawing numbers (and other information required for exact identification of location for each item), manufacturer and brand, ASTM or Federal Specification Number (if any) and such additional information as may be required by specifications for particular item being furnished. In addition, catalogs must be marked to indicate specific items submitted for approval.
    - 1. A copy of letter must be enclosed with items, and any items received without identification letter will be considered "unclaimed goods" and held for a limited time only.
    - 2. Each sample, certificate, manufacturers' literature and data must be labeled to indicate the name and location of the Cemetery, name of Contractor, contract number, manufacturer, brand, specification section number, ASTM (as applicable), and location(s) on project.
    - 3. Required certificates must be signed by an authorized representative of manufacturer or supplier of material, and by Contractor.
  - C. In addition to complying with the applicable requirements specified in preceding Article 1.9, samples which are required to have Laboratory Tests under the separate sections of the specification must be tested, at the expense of Contractor, in a commercial laboratory approved by Contracting Officer.
    - 1. Laboratory must furnish Contracting Officer with a certificate stating that it is fully equipped and qualified to perform intended work, is fully acquainted with specification requirements and intended use of materials and is an independent establishment in no way connected with organization of Contractor or with manufacturer or supplier of materials to be tested.
    - 2. Certificates must also set forth a list of comparable projects upon which laboratory have performed similar functions during past five years.
    - 3. Samples and laboratory tests must be sent directly to an approved commercial testing laboratory.
    - 4. Contractor must send a copy of transmittal letter to the Contracting Officer with submission of material to a commercial testing laboratory.

5. Laboratory test reports must be sent directly to Contracting Officer for appropriate action.
  6. Laboratory reports must list contract specification test requirements and a comparative list of the laboratory test results. When tests show that the material meets specification requirements, the laboratory must so certify on test report.
  7. Laboratory test reports must also include a recommendation for approval or disapproval of tested item.
- D. If submittal samples have been disapproved, resubmit new samples as soon as possible after notification of disapproval. Such new samples must be marked "Resubmitted Sample" in addition to containing other previously specified information required on label and in transmittal letter.
- E. Approved samples will be kept on file by the Contracting Officer at the site until completion of contract, at which time such samples will be delivered to Contractor as Contractor's property. Where noted in technical sections of specifications, approved samples in good condition may be used in their proper locations in contract work. At completion of contract, samples that are not approved will be returned to Contractor only upon request and at Contractor's expense. Such request should be made prior to completion of the contract. Disapproved samples that are not requested for return by Contractor will be discarded after completion of contract.
- F. Submittal drawings (shop, erection or setting drawings) and schedules, required for work of various trades, must be checked before submission by technically qualified employees of Contractor for accuracy, completeness and compliance with contract requirements. These drawings and schedules must be stamped and signed by Contractor certifying to such check.
1. For each drawing required, submit one Portable Document Format (PDF) drawing file.
  2. Each drawing must have marked thereon, proper descriptive title, including Cemetery location, project number, manufacturer's number, reference to contract drawing number, detail Section Number, and Specification Section Number.
  3. One electronic file of approved or disapproved shop drawings will be forwarded to Contractor.
  4. When work is directly related and involves more than one trade, shop drawings must be submitted to Contracting Officer under one cover.
  5. Drawing files must be emailed to the CO, COR and Project Manager. (Files attachments are not to exceed 10MB per email message.)
- 1.11 Samples, shop drawings, test reports, certificates and manufacturers' literature and data, shall be submitted for approval to the CO, COR and Project Manager:

|                                         |                   |                                    |
|-----------------------------------------|-------------------|------------------------------------|
| Shane Hilts<br>Contracting Officer (CO) | Cemetery Director | John Roark<br>Project Manager (PM) |
|-----------------------------------------|-------------------|------------------------------------|

|                                                                                                                                   |                                                                                                                                         |                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| National Cemetery<br>Administration<br>18434 Joplin Rd,<br>Triangle, VA 22172 Tel:<br>(402) 714-8099 Email:<br>shane.hilts@va.gov | Fort Snelling National<br>Cemetery<br><br>7601 34th Ave S,<br>Minneapolis, MN 55450<br>Tel: <a href="tel:6127261127">(612) 726-1127</a> | National Cemetery<br>Administration<br><br>Tel.: 317.447.1140<br>Email: john.roark2@va.gov |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|

- 1.12 At the time of transmittal, the Contractor must also send a copy of the complete submittal directly to the Contracting Officer.
- 1.13 One copy of the approved submittals must be sent directly to the CO and COR.

### END OF SECTION

## **SECTION 01 45 29 - TESTING LABORATORY SERVICES**

### **PART 1 - GENERAL**

#### **1.1 DESCRIPTION:**

This section specifies materials testing activities and inspection services required during project construction to be provided by a Testing Laboratory retained and paid for by Contractor.

#### **1.2 APPLICABLE PUBLICATIONS:**

- A. The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.
- B. American Association of State Highway and Transportation Officials (AASHTO):
- T27-99.....Standard Specification for Sieve Analysis of Fine and Coarse Aggregates
- T96-02.....Standard Specification for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
- T99-01 .....Standard Specification for the Moisture-Density Relations of Soils Using a 2.5 Kg (5.5 lb.) Rammer and a 305 mm (12 in.) Drop

- T104-99.....Standard Specification for Soundness of Aggregate  
by Use of Sodium Sulfate or Magnesium Sulfate
- T180-01.....Standard Specification for Moisture-Density  
Relations of Soils using a 4.54 kg (10 lb.) Rammer  
and a 457 mm (18 in.) Drop
- T191-02.....Standard Specification for Density of Soil In-Place  
by the Sand-Cone Method
- T205-86(R1996) .....Standard Specification for Density of Soil In-Place  
by the Rubber-Balloon Method
- T238-97.....Standard Specification for Density of Soil and Soil-  
Aggregate in Place by Nuclear Methods (Shallow  
Depth)
- C. American Society for Testing and Materials (ASTM):
- C 88-99a.....Standard Test Method for Soundness of Aggregates  
by Use of Sodium Sulphate or Magnesium Sulfate.
- C 131-03.....Standard Test Method for Resistance to  
Degradation of Small-Size Coarse Aggregate by  
Abrasion and Impact in the Los Angeles Machine.
- C 136-96a.....Standard Test Method for Sieve Analysis of Fine  
and Coarse Aggregates.
- D 75.-97 .....Standard Practice for Sampling Aggregates.
- D698 (2012e1) .....Standard Test Method for Laboratory Compaction  
Characteristics of Soil Using Standard Effort
- D 854-02 .....Standard Test Method for Specific Gravity of Soil  
Solids by Water Pycnometer.
- D1556-00 .....Standard Test Method for Density and Unit Weight  
of Soil in Place by the Sand-Cone Method
- D1557-12 .....Test Method for Laboratory Compaction  
Characteristics of Soil Using Modified Effort

|                       |                                                                                                                                                                           |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D2167-94(R2001) ..... | Standard Test Method for Density and Unit Weight<br>of Soil in Place by the Rubber Balloon Method                                                                         |
| D2216-98 .....        | Standard Test Method for Laboratory<br>Determination of Water (Moisture) Content of Soil<br>and Rock by Mass                                                              |
| D2974-(2000).....     | Standard Test Methods for Moisture, Ash, and<br>Organic Matter of Peat and Other Organic Soils                                                                            |
| D3740-(2001).....     | Standard Practice for Minimum Requirements for<br>Agencies Engaged in the Testing and/or Inspection<br>of Soil and Rock as Used in Engineering Design<br>and Construction |

### **1.3 REQUIREMENTS:**

- A. Accreditation Requirements: Testing Laboratory retained and paid for by the Contractor must be accredited by one or more of the National Voluntary Laboratory Accreditation Program (NVLAP) programs acceptable in the geographic region for the project. Furnish to the Contracting Officer a copy of the Certificate of Accreditation and Scope of Accreditation. For testing laboratories that have not yet obtained accreditation by a NVLAP program, submit an acknowledgement letter from one of the laboratory accreditation authorities indicating that the application for accreditation has been received and the accreditation process has started, and submit to the Contracting Officer for approval, certified statements, signed by an official of the testing laboratory attesting that the proposed laboratory, meets or conforms to the ASTM standards listed below as appropriate to the testing field.
  - 1. Laboratories engaged in testing of construction materials shall meet the requirements of ASTM E329.
  - 2. Laboratories engaged in testing of soil and rock, as used in engineering design and construction, shall meet the requirements of ASTM D3740.
  - 3. Laboratories engaged in non-destructive testing (NDT) shall meet the requirements of ASTM E543.
  - 4. Laboratories engaged in Hazardous Materials Testing shall meet the requirements of OSHA and EPA.
- B. Inspection and Testing: Testing laboratory must inspect materials and workmanship and perform tests described herein and additional tests requested by Contracting Officer. When it appears materials furnished, or work performed by Contractor fails to meet construction contract requirements, Testing Laboratory must direct attention of Contracting Officer to such failure.

- C. Written Reports: Testing laboratory must submit test reports to Contracting Officer, Contractor, and Local Building Authority within 24 hours after each test is completed unless other arrangements are agreed to in writing by the Contracting Officer. Submit reports of tests that fail to meet construction contract requirements on colored paper.
- D. Verbal Reports: Give verbal notification to Contracting Officer immediately of any irregularity.

## **PART 2 - PRODUCTS (NOT USED)**

## **PART 3 - EXECUTION**

### **3.1 EARTHWORK TESTING:**

- A. General: The Testing Laboratory must provide qualified personnel, materials, equipment, and transportation as required to perform the services identified/required herein, within the agreed to schedule and/or time frame. The work to be performed shall be as identified herein and must include but not be limited to the following:
  - 1. Observe fill and subgrades during proof-rolling to evaluate suitability of surface material to receive fill or base course. Provide recommendations to the Contracting Officer regarding suitability or unsuitability of areas where proof-rolling was observed. Where unsuitable results are observed, witness excavation of unsuitable material and recommend to Contracting Officer extent of removal and replacement of unsuitable materials and observe proof-rolling of replaced areas until satisfactory results are obtained.
  - 2. Provide part time observation of fill placement and compaction and field density testing in work areas and provide part time observation of fill placement and compaction and field density testing in geogrid Flat Marker headstone support system areas to verify that earthwork compaction obtained is in accordance with these specifications.
- B. Compaction Testing:
  - 1. Determine maximum density and optimum moisture content for each type of fill, backfill, subgrade and topsoil material used, in compliance with ASTM D698.
    - a. Subgrade: At least 1 test for every 1850 sq-m (20,000 sq.-ft.) of compacted backfill, but in no case fewer than 10 tests.
    - b. Topsoil Backfill: At least 1 test for every 1850 sq-m (20,000 sq.-ft.) of compacted topsoil backfill, but in no case fewer than 10 tests.
  - 2. Determine maximum density and optimum moisture content for each type of for aggregate base material used, in compliance with ASTM D1557.

- a. Aggregate Base: At least 1 test for every 185 sq-m (2,000 sq.-ft.) of compacted aggregate base, but in no case fewer than 10 tests.
  3. Field density tests: Testing agency will test compaction of soils in place according to ASTM D 1556, ASTM D 2167, and ASTM D 2937, as applicable.
  4. Should the testing laboratory propose alternative testing methods, they should provide satisfactory explanation to the Contracting Officer before the tests are conducted.
  5. When testing agency reports that subgrades, fills, or backfills have not achieved degree of compaction specified, scarify and moisten or aerate, or remove and replace soil to depth required; recompact and retest until specified compaction is obtained.
- C. Materials Testing:
1. Laboratory Testing: Materials Testing is to be conducted by an independent laboratory, recognized by the State of Minnesota, with the experience and capability to conduct the testing indicated and that specializes in types of tests to be performed.
  2. Provide manufacturer's laboratory tests analyses with product submittals indicating compliance with these specifications.
  3. On-Site Existing Topsoil Testing: Test existing topsoil in each region as indicated in these specifications. (See Turf Renovation.)
  4. Imported Topsoil Testing: Test each imported topsoil as indicated in these specifications. (See Turf Renovation.)
  5. Post-Amendment Topsoil Testing: Test each amended topsoil as indicated in these specifications. Test results must indicate that the amended topsoil is not deficient in essential nutrients and meets the requirements of these specifications. (See Turf Renovation.)
  6. Compost Testing: Provide Certificates showing compliance with these specifications. (See Turf Renovation.)
  7. Elemental Sulfur: Provide Manufacturer's Material Data Sheets showing compliance with these specifications. (See Turf Renovation.)
  8. Agricultural Gypsum: Provide Manufacturer's Material Data Sheets showing compliance with these specifications. (See Turf Renovation.)
  9. Mycorrhizae: Provide Manufacturer's Material Data Sheets showing compliance with these specifications. (See Turf Renovation.)

## **END OF SECTION**

### **SECTION 02 21 13 - SITE SURVEYS (LIMITED)**

- 1.1 GENERAL:** The contractor must perform all surveying services necessary to provide a headstone location/elevation and topographic survey of existing site features. This survey will form the basis for developing the final grade for re-

installing the headstones at the proper elevations in order to achieve proper headstone alignment, and positive drainage. Provide a legend on each sheet identifying all features on the drawings. Provide north arrow and graphic scale. Show headstone direction arrows for each section which indicate the direction the front of the marker is facing the viewer. Provide burial section data in a table showing section number, gravesite grid size, headstone type (upright or flat), number of headstones, number of reserved gravesites, area of sod in square-feet in each section.

- 1.2 PROJECT LIMITS:** The area to be surveyed must include the sections or portions thereof, at the National Cemetery that are to be renovated as part of this contract. Unless indicated on the plan or approved by the Contracting Officer (CO), the survey limits must extend to a minimum of 20 feet beyond the sections being renovated. The survey must locate each headstone using standard survey coordinates and record the position and elevation, and the contour of the grades.
- 1.3 HEADSTONE SURVEY:** The contractor must survey the same corner (i.e., upper left) of each marker at finished grade, taken while facing the front of the marker. It will be this corner that will be used to realign the headstones. Once selected, the selected corner must not change throughout the contract. The survey must locate each marker, using standard survey coordinates and map them in CAD format drawings.
  - A. HEADSTONE REALIGNMENT PLANS:** Overlay a proposed realignment grid that is aligned with as many of the marker coordinates as possible. The proposed realignment grid must align with all adjacent realignment grids wherever possible so that headstones in adjacent sections visually align with each other. The realignment grid is based on the gravesite dimensions for the section under renovation, (i.e., 5'x10', 4'x8', 3'x8', 4'x4', 3'x3'). Highlight headstones whose existing location deviates more than 6 inches from the proposed realignment grid. Tie the realignment grids with dimensions to horizontal controls such as existing gravesite control monuments (if exist), other permanent existing survey datum, or temporary Contractor-installed gravesite control markers at both ends of each row of headstones. Show the locations of temporary control markers on the Headstone Realignment Plans. Submit 2 hard copies stamped "PRELIMINARY – FOR REVIEW" and PDF format electronic files to the Midwest District Project Manager and COR for review. **NO MARKERS WILL BE MOVED FROM THEIR ORIGINAL LOCATION UNTIL THE PROPOSED REALIGNMENT HAS BEEN SUBMITTED AND APPROVED BY THE COR.**
  - B. APPLICABLE PUBLICATIONS**
    - A.** Publications listed below form a part of this specification to extent referenced. Publications are referenced in text by basic designation only.

B. American Society for Testing and Materials (ASTM):

- D698-00 .....Laboratory Compaction Characteristics of Soil  
Using Standard Effort (12,400 ft. lbf/ft<sup>3</sup> (600 kN  
m/m<sup>3</sup>))
- D1556-00 .....Density and Unit Weight of Soil in Place by the  
Sand-Cone Method
- D1557-12 .....Laboratory Compaction Characteristics of Soil  
Using Modified Effort (56,000 ft-lbf/ft<sup>3</sup> (2700 kN  
m/m<sup>3</sup>))
- D2940-98 .....Graded Aggregate Material for Bases or Subbases  
for Highways or Airports
- Division 300.....State of Kansas Department of Transportation  
(NMDOT) Division 300 - (2019) Standard  
Specifications - Section 303 – Base Course

**C. UTILIZATION OF EXCAVATED MATERIALS**

- A. Contractor excavated and stockpiled materials may be used as fill and backfill materials as specified. Excavated materials must be stockpiled in designated stockpile areas. No excavated material must be disposed of to obstruct the flow of any stream or road, endanger a partly finished structure, impair the efficiency or appearance of any structure, or be detrimental to the completed work in any way.
- B. Excess soil material shall be disposed of off-site in a lawful manner.

**PART 2 - PRODUCTS – (Not Used)**

**PART 3 - EXECUTION**

**3.1 HEADSTONE SURVEY, TOPOGRAPHIC SURVEY, and GRADING PLAN:**

- A. New Headstone Elevations and Finished Grading Plan: Based on the completed survey of existing headstone elevations and their location, and the topographic survey, the Contractor must develop and submit for approval a final headstone elevation/position plan and section grading plan showing spot elevations at proposed finish grade for each headstone and proposed contours at 1/10 foot contour intervals. Show existing 1-foot contours for reference. The finished grading plan shall incorporate the addition of topsoil to existing grades as

indicated on the drawings to fill low spots and depressions in the burial sections. Proposed finished grades must provide positive drainage off burial sections and into drainage structures where applicable. (See Exhibit 'C'.) Proposed Grading plans showing excessive cut or fill at existing grades will not be approved.

- B. Contour Interval: Existing contours shall be indicated by broken lines, drawn at 1/10 foot intervals, or otherwise approved or required interval. Proposed contours shall be indicated by solid lines drawn at 1/10 foot intervals. One-foot interval contours shall be drawn with a heavier line weight than 1/10-foot contour lines. Label elevation of each one-foot contour interval.
- C. Spot Elevations: Spot elevations shall be provided on an approximate 25 foot minimum grid pattern, and at all other appropriate locations where site features, structures, paved surfaces, and utilities require greater detail to illustrate existing conditions. Provide spot elevations at the top and bottom of abrupt changes in grade. Provide spot elevations at headstones and rim elevations for drain inlets. Provide spot elevations at trees.

## **END OF SECTION**

### **SECTION 31 00 00 - EARTHWORK**

#### **PART 1 - GENERAL**

##### **D. DESCRIPTION OF WORK**

- A. This section includes the requirements for earthwork including, but not limited to, the following:
  - 1. Vegetation removal.
  - 2. Site preparation.
  - 3. Excavation and embankment.
  - 4. Filling and backfilling.
  - 5. Site grading.
  - 6. Base aggregate placement and compaction.
  - 7. Levelling sand.
  - 8. Amended topsoil placement and compaction.
  - 9. Soil disposal.

##### **E. DEFINITIONS**

- A. Headstones: The term "Headstone" and "Marker" when used generically, includes both upright headstones and flat marker headstones. Work in this project includes both Upright and Flat Marker Headstones. When there is need to distinguish between the two types of headstones, Upright Headstones will be referred to as "Upright Headstones" or "Uprights"; and Flat Marker Headstones will be referred to as "Flat Markers" or "Flats".

- B. Borrow Material: Borrow materials are soils generated during excavation operations at the site.
  - 1. Backfill: Backfill shall be obtained from onsite excavation. The material shall be used as indicated on the Drawings. It must be free of organic matter, debris and other deleterious substances. The material shall contain no particles greater than 2 inches in any dimension.
- C. Unsatisfactory Materials: Materials that do not comply with the requirements listed above are unsatisfactory. Unsatisfactory materials also include nonengineered fills; trash; refuse; backfills from previous construction; and material classified as satisfactory that contain or are contaminated with unsatisfactory material. The Contracting Officer must be notified of any contaminated materials.
- D. Degree of compaction: Degree of compaction is expressed as a percentage of the maximum density obtained by the test procedure presented in ASTM D 698, abbreviated as a percent of laboratory maximum density.
- E. Unauthorized excavation: Removal of materials beyond indicated sub-grade elevations or dimensions without written authorization by the Contracting Officer. No payment will be made for unauthorized excavation or remedial work required to correct unauthorized excavation.
- F. Subgrade: Surface or elevation remaining after completing excavation, or top surface of a fill or backfill immediately below base or topsoil materials.
- G. Topsoil: ASTM D 5268; natural or cultivated surface-soil layer containing organic matter and sand, silt, and clay particles, conforming to USDA classification for Loam or Sandy Loam; friable, pervious, and compatible in texture with the existing soil to prevent a perched water table; a minimum of 4SAR percent and a maximum of 6 percent organic material content; pH range between 6 and 8; reasonably free of subsoil, clay lumps, gravel, rocks and other objects more than 13 mm (1/2-inch) in any dimension; and free of weeds, roots, and other extraneous materials harmful to plant growth.
- H. Amended Topsoil: (See Section 32 92 24, "Turf Renovation".)
- I. Abbreviations:
  - 1. USDA. United States Department of Agriculture.
  - 2. ASTM. American Society for Testing and Materials.
  - 3. NTEP. National Turfgrass Evaluation Program

## **F. RELATED WORK**

- A. Protection of existing utilities, fire protection services, existing equipment, roads, and pavements: Section 01010, GENERAL REQUIREMENTS.

- B. Vegetation removal: Section 32 92 24, TURF RENOVATION.
- C. Amendment of existing site topsoils: Section 32 92 24, TURF RENOVATION.

## **G. SUBMITTALS**

- A. Unless otherwise noted, submittals shall be made a minimum of 14 days before commencing the Work specified in this Section. The following shall be submitted in accordance with Section 01 33 23 - SHOP DRAWINGS, PRODUCT DATA, and SAMPLES.
  - 1. Import Materials: Contractor must submit the following for each imported material a minimum of 14 days prior to delivery:
    - a. Material source(s).
    - b. Particle size analysis in accordance with ASTM C136.
    - c. Material samples.
  - 2. Import Topsoil Testing: Contractor must submit Agricultural Soil Suitability Test and Report with recommendations for correcting soil deficiencies for review and approval of COR for each source of import topsoil supplied to the project.
  - 3. Compost Analysis: Provide documentation from supplier that compost meets criteria below in Part 2—Products. Supplier(s) shall provide testing from an independent soil testing laboratory to test representative sample(s) of compost and furnish compost analysis report for approval of Midwest District Agronomist through the COR for review and approval.
  - 4. Equipment List: Contractor must submit a list of equipment to be utilized for the work 7 days prior to mobilization. The list must include equipment make, model, year, tire or track dimensions, weight and other information. The use of wheeled equipment is prohibited on cemetery burial sections. Contractor must use only track-type equipment to prevent overcompaction of soils.
  - 5. Construction Procedures Plan: Contractor must submit a plan 14 days minimum prior to material construction that includes, but not be limited to: marker surveying and identification/tagging; marker pulling (describe method how work will be accomplished without removing markers from the gravesites); removing irrigation heads, capping irrigation risers; excavation and embankment; rough grading; cross-ripping subgrade; amending subsoil; Flat Marker headstone support system installation; marker re-installation (both uprights and flats); re-installing irrigation heads and restoring irrigation system; placing and amending topsoil; sod installation; equipment use; erosion control; and protection to be provided in the event of rain, wind, heat or other potential cause of damage.

## **H. APPLICABLE PUBLICATIONS**

- C. Publications listed below form a part of this specification to extent referenced. Publications are referenced in text by basic designation only.

- D. American Society for Testing and Materials (ASTM):

D698-00 .....Laboratory Compaction Characteristics of Soil  
Using Standard Effort (12,400 ft. lbf/ft<sup>3</sup> (600 kN  
m/m<sup>3</sup>))

D1556-00 .....Density and Unit Weight of Soil in Place by the  
Sand-Cone Method

D1557-12 .....Laboratory Compaction Characteristics of Soil  
Using Modified Effort (56,000 ft-lbf/ft<sup>3</sup> (2700 kN  
m/m<sup>3</sup>))

D2940-98 .....Graded Aggregate Material for Bases or Subbases  
for Highways or Airports

Division 300.....State of Minnesota Department of Transportation  
(NMDOT) Division 300 - (2019) Standard  
Specifications - Section 303 – Base Course

## **I. UTILIZATION OF EXCAVATED MATERIALS**

- C. Contractor excavated and stockpiled materials may be used as fill and backfill materials as specified. Excavated materials must be stockpiled in designated stockpile areas. No excavated material must be disposed of to obstruct the flow of any stream or road, endanger a partly finished structure, impair the efficiency or appearance of any structure, or be detrimental to the completed work in any way.

- D. Excess soil material shall be disposed of off-site in a lawful manner.

## **PART 2 – PRODUCTS**

### **2.1 MATERIALS**

- A. Aggregate Base/Backfill for supporting and backfilling around Upright Headstones must be 3/8-inch minus crushed aggregates in conformance to Minnesota DOT (NMDOT) Standard Specifications, Section 519, Subsection 519.2.2, Fine Aggregates, Grading No. 2.

| Sieve Size % Passing by Weight |                      |
|--------------------------------|----------------------|
| Sieve Size                     | % of Passing         |
| 3/4 in.                        | --                   |
| 1/2 in.                        | 100                  |
| 3/8 in.                        | 90-100               |
| No. 4                          | 70-85                |
| No. 8                          | 50-70                |
| No. 16                         | 35 to 55             |
| No. 30                         | 20 to 35             |
| No. 50                         | 8 to 20              |
| No. 100                        | 2 to 10              |
| No. 200                        | sand equivalent = 75 |

- B. Base Aggregate for Flat Marker geogrid support system must be 3/4 - 0 inch crushed aggregate in conformance to NMDOT (3/4-inch minus crushed aggregate).
- C. Leveling Sand for Flat Markers must be 1/4-inch minus crushed aggregate with 90-100 percent passing the No. 4 screen and maximum 6 percent passing the No. 200 screen
- D. Amended Topsoil: (See “Turf Renovation”).)
- E. Elemental Sulfur: Granular, biodegradable, containing a minimum of 90 percent sulfur, with a minimum of 99 percent passing through No. 6 sieve and a maximum of 10 percent passing through No. 40 sieve.
- F. Agricultural Lime: Pelletized Lime (90% Calcium Carbonate equivalent derived from ground limestone).
- G. Gypsum Equivalent: Mined Anhydrite Calcium Sulfate and Hydrated Calcium Sulfate. Minimum 90 percent calcium sulfate, pelletized, SGN 250.
  - a. Available Product: ‘Best’, Soil Buster; or approved equal.
- H. Organic Soil Amendments: Acceptable types of organic soil amendments include peat moss, humus or peat, well-rotted manure, various mature composts, and commercially available combinations thereof. Acceptable compost may be derived from natural organic sources such as food or animal residuals, yard trimmings, or biosolids. Organic soil amendment shall be free of all woody fibers, seeds, and leaf structures, plastic and other petroleum products, free of toxic and non-organic matter, and free from objectionable odors. Unacceptable sources of organic matter include untreated sludge from wastewater treatment plants, fresh manure, sawdust, and immature composts.

- I. Compost: Well-decomposed, commercially manufactured, stable, and weed-free organic matter from agricultural, food, biosolids, or yard debris sources; free of refuse (less than 1 percent by dry weight), plastics, contaminants or any material toxic to plant growth; composted for a minimum of 9 months and reached a monitored temperature of 140 degrees F for at least one week; tested, at minimum, every six months for noxious weeds; and conforming to the following:
  1. Organic Matter Content: 60 to 80 percent of dry weight as determined by ash method.
  2. Moisture Content: 35 to 55 percent by weight.
  3. Inerts: less than 0.5% by dry weight.
  4. pH range of 6.0 to 8.0.
  5. Soluble salts: less than 2 decisiemens (dS)/m.
  6. Particle size: 100 percent passing through 1/2-inch (13-mm) sieve.
  7. Nutrient Content: sum of N, P, K percent of wet weight, less than 5%.
  8. Trace Metals: Pass per US EPA Class A Standard, 40 CFR 503.32 (a).
  9. Maturity Indicator (bio-assay): Greater than 80% seedling emergence relative to positive control.
  10. Stability Indicator: Carbon dioxide evolution rate is less than 8 mg CO<sub>2</sub>-C per g OM per day – “Stable”.
  11. Tested, at minimum, every six months for noxious weeds.
  12. Organic matter source (feedstock): Agricultural, food, or industrial residuals; biosolids; yard trimmings; or source-separated or compostable mixed solid waste.
  13. Organic Matter Content: 30 to 65 percent of dry weight as determined by ash method.
  14. Processed to meet U.S. Composting Council's Seal of Testing Assurance Program, or equivalent.
- J. Peat Moss: Peat shall be a natural product of peat moss, hypnum moss, reed sedge peat/peat humus derived from a fresh-water site. Peat shall be shredded and granulated to pass through a 13 mm (1/2 inch) mesh screen and conditioned in storage piles for at least six months after excavation. Characteristics as follows: pH (3.5 to 4.5); EC (0-0.1 mmhos/cm); SAR <5; Dry Density (60-90 g/l); Moisture content (38 to 50%); Organic matter (98-99%).
- K. Imported Topsoil: ASTM D5268; natural or cultivated surface-soil layer having the following characteristics:
  1. Sand, silt, and clay particles in percentages conforming to USDA classification for Loam or Sandy Loam;
  2. Micro and macro-nutrients in proper quantities essential for plant health;
  3. Contain a minimum of 4 percent and a maximum of 10 percent organic material content;
  4. Have pH range between 6.5 and 7.5;
  5. Be friable, pervious, and black or a darker shade of brown;
  6. Be free of subsoil, clay lumps, gravel, rocks and other objects more than 1/2 inch (13mm) in any dimension;

7. Be free of weeds, roots, and other extraneous materials harmful to plant growth.
- L. Mycorrhizae: Endomycorrhizal powder inoculum consisting of the following 4 species blend of propagules of arbuscular mycorrhizal fungi: Glomus intraradices, Glomus mosseae, Glomus aggregatum, and Glomus etunicatum. Minimum 100,000 spores/propagules per pound. The powder particle size shall be less than 300 microns (100 percent passing the #50 screen). Available Products: 'MycoApply Micronized Endo' by Mycorrhizal Applications, Inc.; 'Super Fine Endo' by Mycorrhizal Products.com; or equal.

## **PART 3 – EXECUTION**

### **3.1 SITE PREPARATION**

- A. Lines and Grades: With the services of a Registered Professional Land Surveyor or Registered Civil Engineer, specified in Site Survey section, the Contractor must survey and document existing lines and grades and submit for approval, new lines and grades in order to achieve grades that provide a smooth surface, free from irregular surface changes and provide positive drainage. The intent is to eliminate any irregular surface changes and depressions that trap surface water and provide movement of water off cemetery sections and/or into storm water structures. Grading must comply with compaction requirements and grade cross sections, lines, and elevations indicated by the approved Contractor-provided Grading Plan. Where spot grades are indicated the grade must be established based on interpolation of the elevations between the spot grades while maintaining appropriate transition at structures and paving and uninterrupted drainage flow into inlets.
- B. Trees and Shrubs: Protect from damage, existing trees and shrubs which are not shown to be removed in construction area. Immediately repair damage to existing trees and shrubs by trimming, cleaning and painting damaged areas, including roots, in accordance with standard industry horticultural practice for the geographic area and plant species. Do not store materials closer to trees and shrubs that are to remain, than the farthest extension of their limbs. If the Contractor is obstructed by tree roots within the drip-line of a standing tree that are 2 inches in diameter or greater, the Contractor must request an inspection by the COR. The COR will inspect and determine if any roots will be pruned.
- C. Vegetation Removal: Remove turfgrass sod, weeds and their roots from within limit of work area. Cut a clean line between sod to be removed and sod in adjacent cemetery sections to be protected, in addition to any sod in adjacent cemetery sections requiring removal to accomplish excavation for work required. Haul vegetation off-site in a lawful manner.

### **3.2 SUBGRADE PREPARATION**

- A. The ground surface on which fill is to be placed must be cleared of vegetation.

- B. After vegetation removal, shape the subgrade to lines, grades, and sections shown on the approved, Contractor-provided grading plan, providing allowance for incorporation of soil amendments and placement of 2-inches of imported topsoil.
- C. Soft or otherwise unsatisfactory material must be removed and replaced with satisfactory excavated material or other approved material as directed. Existing low areas and those resulting from removal of unsatisfactory material must be brought up to required grade with satisfactory materials, and the entire subgrade must be shaped and compacted as specified.
- D. Stones 2 inches and larger must be removed and replaced with satisfactory excavated material or other approved material as directed.
- E. If the Contractor excavates below the lines and grades indicated on the Construction Drawings, Contractor must place fill to elevate these areas back to grade at no additional cost to the Government.
- F. The prepared subgrade surface must be reasonably smooth, free of holes, depressions greater than 3 inch deep, or protrusion extending above the surface more than 3 inch. No overlying materials shall be placed until the subgrade has been checked and approved. The subgrade surface must be protected and restored if damaged.

### **3.3 SUBGRADE AMENDMENT**

- A. Prior to compacting final subgrades, cross-rip existing subsoil to a depth of 8 to 12-inches to scarify, and amend with gypsum equivalent and organic soil amendments.
- B. Broadcast gypsum equivalent at a rate of 1 ton per acre (~50 lbs./1000 sq-ft); and spread approved organic soil amendments throughout the turf renovation area. The rate of incorporation of the organic soil amendments shall be based upon the organic content of the tested and approved organic soil amendments, so as to produce a finished soil with an organic matter content of between four (4%) and six percent (6%).
- C. Incorporate the gypsum equivalent and organic soil amendment by disc harrowing, rototilling or other means to provide a uniformly homogenous mixture to depth of 6 to 8 inches.
- D. Supply additional organic soil amendment material, after in-place testing and approval, as may be needed to give the required organic matter content and finished grades under the Contract without additional cost to the Government.

### **3.4 SOIL MOISTURE CONTROL**

- A. Uniformly moisten or aerate subgrade and each subsequent fill or backfill soil layer before compaction to within 2 percent of optimum moisture content.
  - 1. Do not place backfill or fill soil material on surfaces that are muddy, frozen, or contain frost or ice.
  - 2. Remove and replace, or scarify and air dry otherwise satisfactory soil material that exceeds optimum moisture content by 2 percent and is too wet to compact to specified dry unit weight.

### **3.5 FILL AND BACKFILL**

- A. Place backfill and fill soil materials in layers not more than 8 inches (200 mm) in loose depth for material compacted by heavy compaction equipment, and not more than 4 inches (100 mm) in loose depth for material compacted by hand-operated tampers.
- B. Place backfill and fill soil materials evenly on all sides of structures to required elevations, and uniformly along the full length of each structure.
- C. Compact soil materials to not less than the following percentages of maximum dry density according to ASTM D 698 for cohesive soils and ASTM D 1557 for granular soils:
  - 1. Under lawn, compact each layer of backfill or fill soil material at 85 percent.
  - 2. Under flat headstones, compact each layer of backfill or fill soil material at 95 percent.
  - 3. Under Flat Marker headstone support system compact each layer of backfill or fill soil material at 95 percent.

### **3.6 GRADING**

- A. Grading must be in conformance with the approved, Contractor-provided, Site Grading Plan. Excavation and finish grades must be constructed to within plus or minus 0.1 foot of the indicated grades. The finished surface of the excavations and fills must be free of depressions and must be reasonably smooth in accordance with the grade tolerances.
- B. Establish rough grades per the approved, Contractor-provided Site Grading Plan, making allowance for the placement of 2-inches of imported topsoil, and headstone support system to depths indicated.
- C. Slope grades to prevent ponding. Finish subgrades to required elevations within the following tolerances:
  - 1. Lawn: Plus or minus 1 inch (25 mm).
  - 2. Flat Markers: Plus or minus 1/2 inch (13 mm).

### **3.7 BASE COURSE (UPRIGHT HEADSTONES)**

- A. Place base course on subgrades free of mud, frost, snow, or ice.
- B. On prepared subgrade, place and compact base course under markers as follows:
  - 1. Place base course 3 inches minimum depth compacted in 1-1/2 inch lifts.
  - 2. Compact base course at optimum moisture content to required grades, lines, cross sections, and thickness to 95 percent of maximum dry density according to ASTM D-1557.
  - 3. See Upright Headstone Installation on Exhibit 'B' – Headstone Setting Details.

### **3.8 AGGREGATE BACKFILL (UPRIGHT HEADSTONES)**

- A. Place aggregate backfill along sides of properly positioned markers as follows:
  - 1. Place and compact aggregate backfill in 1-1/2 inch lifts until within 4 inches of finish grade.
  - 2. Compact backfill at optimum moisture content to required grades, lines, cross sections, and thickness to 95 percent of maximum dry density according to ASTM D 1557.
  - 3. See Upright Headstone Installation on Exhibit 'B' – Headstone Setting Details.

### **3.9 BASE COURSE (FLAT MARKER SUPPORT SYSTEM)**

- A. Place base course for Flat Marker support system on subgrades free of mud, frost, snow, or ice.
- B. On prepared subgrade, place and compact base course in trench under flat marker support geogrids at flat marker locations as follows:
  - 1. Place base course 6 inches (150 mm) or less in compacted thickness in a single layer.
  - 2. Compact base course at optimum moisture content to required grades, lines, cross sections, and thickness to 95 percent of maximum dry density according to ASTM D 1557.
  - 3. Aggregate finish grades must be constructed to within plus or minus 0.1 foot of the indicated grades. The finished surface of the aggregate must be free of depressions and must be reasonably smooth in accordance with the grade tolerance.
  - 4. See Flat Marker Installation Details on Exhibit 'B' – Headstone Setting Details.

### **3.10 TOPSOIL PLACEMENT**

- A. Placement and compaction of imported topsoil shall occur after the initial setting of the Flat Marker headstones onto the plastic headstone support system to prevent intermingling of subgrade soil and imported topsoil during excavation for placement of support system aggregate base. (See "Section 32 39 20 – FLAT

MARKER SUPPORT SYSTEM, HEADSTONE SETTING, ALIGNMENT, BACKFILLING, AND HEADSTONE CLEANING”).)

- B. Scarification: Scarify previously compacted subgrade to a depth of 1 to 2 inches prior to placement of imported topsoil to prevent soil layering.
- C. Mycorrhizae: Immediately prior to the placement of imported topsoil, broadcast Mycorrhizae onto the scarified subsoil at the 0.5 lb./1000 sq-ft using properly calibrating spreading equipment.
- D. Place and compact 2-inches of imported topsoil throughout the entire area of turf renovation and compact to 85 percent maximum dry density per ASTM D-698. Test compaction levels of topsoil per “SECTION 01 45 29 – TESTING LABORATORY SERVICES.”
  - 1. Compact topsoil using hand or mechanized equipment well suited to soil being compacted. Do not operate heavy mechanized or vibratory compaction equipment which may over-compact the soil. The Contractor must not operate trucks, tractors, and other heavy wheeled equipment in excess of 2 tons, or track equipment in excess of 3 tons on any burial section except when authorized in writing by the COR(s). Topsoil must be thoroughly tamped to prevent further sinkage, but the soil must not be compacted to the degree that it prevents the growth of healthy turf.
- E. Amend topsoil as recommended by results of imported topsoil soil test reports as indicated in “SECTION 32 94 00 – TURF RENOVATION.”

**3.11 FIELD QUALITY CONTROL**

- A. Test compaction of soils in place per “C. SECTION 01 45 29 – TESTING LABORATORY SERVICES.”

**3.12 DISPOSAL OF MATERIALS**

- A. Excess Soil: Excess soil and amended topsoil shall remain the property of the Government. Haul excess soil to location on the cemetery site as directed by the COR and stockpile or spread evenly over the designated area.
- B. Excess soil amendments shall remain the property of the Government. Haul excess materials to a location on the cemetery site as directed by the COR and stockpile in mounds not exceeding 6 feet in height and cover with opaque plastic sheeting as directed by the COR.
- C. Dispose organic matter, vegetation, unsuitable soil materials and excess excavated soil materials generated by the Contractor off-site in a lawful manner.

### **3.13 CLEAN UP**

- A. Upon completion of earthwork operations, clean all work areas within contract limits, and soil manufacturing plant. Remove tools, and equipment. Provide site clear, clean and free of debris. Remove all debris, rubbish, and excess materials unless noted otherwise, from Cemetery Property.

### **END OF SECTION**

## **SECTION 32 39 20 –HEADSTONE SETTING, ALIGNMENT AND CLEANING** (Upright and Flat Marker Headstones)

### **PART 1 - GENERAL**

#### **1.1 SUMMARY**

- A. This section includes:
  - 1. Headstone and gravesite verification survey.
  - 2. Headstone protection and handling.
  - 3. Headstone removal.
  - 4. Flat Marker support system.
  - 5. Headstone setting.
  - 6. Headstone alignment.
  - 7. Headstone backfilling
  - 8. Headstone re-verification.
  - 9. Headstone cleaning.
- B. Related Sections:
  - 1. Division 31 Section "Earthwork" for excavation, filling and backfilling, and rough grading.
  - 2. Division 31 Section "Earthwork" for base aggregates and leveling aggregates.
  - 3. Division 31 Section "Earthwork" for topsoil placement.
  - 4. Division 32 Section "Turf Renovation" for weed control, soil preparation, and sodding.

#### **1.2 SUBMITTALS**

- A. Product Data: For the following:
  - 1. Plastic Flat Marker support geogrid.
  - 2. Headstone Biological Cleaner.

#### **1.3 HEADSTONE and GRAVESITE VERIFICATION**

**TO ENSURE THE ACCURACY OF HEADSTONE AND MARKER PLACEMENT DURING THE RAISE AND REALIGNMENT PROJECT, THE FOLLOWING PROCEDURES WILL BE FOLLOWED BY MEMORIAL SERVICE NETWORK PERSONNEL, CEMETERY PERSONNEL AND**

## **CONTRACTING OFFICER'S REPRESENTATIVE (COR) BEFORE THE RAISE & REALIGNMENT PROCESS BEGINS**

- A. The Midwest District will provide each cemetery with the correct Gravesite Layout Map(s) prior to beginning the Raise & Realign (R&R) project. The Contracting Officer's Representative (COR) or cemetery director/designee as listed in the official contract documents shall conduct an initial gravesite verification survey prior to the (R&R) of any headstone or marker in the Cemetery. The COR will obtain a copy of the Burial Register Report from the Burial Operation Support System (BOSS) and current gravesite layout map(s) obtained from the Midwest District office for the verification survey. The Burial Register Report and gravesite layout map(s) will be used to verify the section, row, grave number, first and last name of each decedent and the condition of the headstone/marker included in the scope of work (SOW). All inconsistencies will be researched, discussed and resolved with the Midwest District Director prior to the R&R of any headstone or marker.
- B. Upon completion of the verification survey and corrective actions as necessary, the COR shall certify the survey by signing the Initial Headstone/Marker Raise & Realign Verification Survey form, (See Attachment 5.) A copy of the signed Headstone/Marker Raise & Realign Verification Survey sheet must be forwarded to the Contracting Officer (CO) which will be incorporated into the contract files; a copy of the survey will be maintained by the cemetery. Additional copies of the survey will be retained by the COR for the official contract files. The CO will give the approval to begin the R&R process.
- C. Prior to the completion of each workday, a Daily Headstone/Marker Raise & Realign Verification Survey will be completed for all headstones and markers raised & realigned, (See Attachment 6). The COR/cemetery director/designee will verify the accuracy of the placement of headstones or markers on the correct gravesite using the same Burial Register Report and gravesite layout map(s) that were used for the initial headstone/marker verification survey. Daily, the COR will inform the contractor/designee of all identified inconsistencies and ensure corrective actions are completed that day. The Daily Headstone/Marker Raise & Realign Verification Survey will be maintained by the COR for certifying payments for completed work, tracking project progress and other administrative needs.
- D. The Contractor's performance and progress on this contract will be measured weekly based on how timely, accurately, and adequately he/she accomplishes and completes the weekly work schedules needed to systematically accomplish the contract work over the duration of the project. In instances where the COR determines that the work is behind schedule, he will notify the Contracting Officer and the Contractor must increase workforce and/or hours of operation at no additional cost to the Government in order to achieve completion of the contract work within the specified timeframe.

## **1.4 HEADSTONE CARE and HANDLING**

- A. HEADSTONES AND/OR MARKERS WILL NOT BE PHYSICALLY REMOVED FROM THE GRAVESITE DURING THE HEADSTONE RAISE AND REALIGNMENT AND TURF RENOVATION PROJECTS.**
- B. When site conditions are restrictive and it is not possible to safely operate equipment while keeping the headstones/markers on the gravesite, the contractor will prepare these sites by hand. At no time may headstones/markers be removed from the gravesites. Markers must never be handled or treated in a disrespectful manner. Do not step on or drive equipment on top of markers. Do not use hand tools to “flip” markers over as a means of relocating markers.
- C. The Contractor is responsible to protect headstones and markers from damage or being discolored from contact with dirt, soil or other contaminants during the course of the contract.

## **PART 2 – PRODUCTS**

### **2.1 PLASTIC HEADSTONE SUPPORT GEOGRID:**

- A. High Density Polyethylene (HDPE) geogrids specifically designed for flat gravesite marker reinforcement and alignment. Geogrids must be in the form of a tray with raised edges to mitigate the shifting markers out of alignment. Geogrid units must be equipped with tabs to interlock with adjoining units.
  - 1. Dimensions - 13” x 13” x 1-5/8”
  - 2. Mass per Piece - 0.53 LBS.
  - 3. Density - 0.0322 LBS. per Cubic Inch.
  - 4. Approved Product: ‘MarkerGrid’, as manufactured by Nursery West Corporation, 11463 Broadacres Road NE, Hubbard, OR 97032;
  - 5. Approved Substitution: ‘GrassPave<sup>2</sup>’ (20m length by 0.5m wide), as manufactured by Invisible Structures, Inc. 1600 Jackson St., Suite 310, Golden, CO 80401; or approved equal.

### **2.2 HEADSTONE CLEANER:**

- A. Non-hazardous, biodegradable liquid that removes stains from mold, algae, mildew, lichens and air pollutants from marble and granite. Contains no acids, salts, or chlorine; pH neutral; safe to use around plants.
- B. Only NCA-approved cleaners shall be used. Cleaners current approved by NCA are ‘D/2 Biological Solution’ by D/2 Biological Solutions, Inc.; and ‘Weather-Zyme’ by United Laboratories, Inc.

### **2.3 AGGREGATES:**

- A. See Section 31 00 00 – EARTHWORK for aggregate bases, aggregate backfill and levelling sand for headstones.

## **PART 3 – EXECUTION**

### **3.1 UPRIGHT MARKER HEADSTONE REMOVAL, RESETTING, REALIGNMENT AND BACKFILL REQUIREMENTS:**

#### **A. WORK OVERVIEW**

1. This subsection outlines Headstone Removal, Resetting, Realignment and Backfill services, defined as: work consisting of extracting Upright headstones; excavating headstone sockets; placement and compaction of foundation aggregate base; resetting, aligning headstones, backfilling and compacting with aggregate backfill, and backfilling and compacting with subbase and/or topsoil at headstones reset in the new earthen sockets, as well as inventorying and assuring accurate placement on each gravesite to be renovated. The Contractor must provide all supervision, professional advice/guidance, labor, parts, materials, equipment, transportation and personnel to provide the services defined herein.
2. Headstones must be accurately adjusted and precisely reinstalled on the correct gravesites by utilizing Contractor-provided temporary gravesite plotting maps, approved, Contractor-provided headstone realignment plans, temporary Contractor-installed gravesite control markers, and temporary Contractor-installed control pin markers at the ends of each gravesite row.

#### **B. UPRIGHT HEADSTONE REMOVAL**

1. Standard marble headstones are approximately 42 inches long, 13 inches wide, 4 inches thick, approximate weight: 230 pounds.
2. All headstones (and other features causing interference with the removal of the Headstones) must be tagged with the unique numbering scheme and surveyed by the contractor prior to removal. Identification tags and survey must be reviewed with the COR prior to headstone removal. Tags must be COR-approved, non-permanent attachments to the Headstones that can be removed without damage to the headstones upon completion of the work.
3. All temporary markers causing interference with the raise and realignment of upright headstone operation must be removed and replaced on the gravesite as soon as practicable, but no later than the close of business each day. Floral, commemorative, or other types of decorations (arrangements)

causing interference with the work must be removed from the gravesite in accordance with the cemetery Floral Policy.

4. Prior to starting work in each burial section, a registered surveyor must install a temporary control pin marker at ends of each gravesite row. The Contractor must create an as-built map showing specific location of each individual headstone by actual number and name in order to facilitate precise headstone reinstallation in the correct location. (See SECTION 02 21 00 - SITE SURVEYS.)
5. Prior to removing headstones from the soil they will be verified with the Contractor-produced gravesite plotting maps. Contractor must verify the accuracy of these maps with the COR prior to removing any headstones.
6. Locations for Reserved Upright Headstones will also be verified with the Contractor-produced grave plotting maps. Contractor must verify the accuracy of these maps with the COR. Contractor must show on the as-built map the specific location of each individual reserved headstone by actual number and name.
7. Headstones will be removed from their sockets by using wooden, and or metal clamps. If metal clamps are used the area that contacts the headstone must be protected with a rigid fabric that will prevent damage to, and marking of, the headstone. Clamps may be attached to a 'Bob Cat' or similar machine to extract headstone from socket. Use care not to scratch or damage headstones in any manner. (Note: In areas where smaller than standard size headstones may occur, coordinate specified measurements and dimensions of required work with COR.)
8. **WHEN HEADSTONES ARE REMOVED FROM THEIR SOCKETS, THEY MUST BE CAREFULLY STORED ON EACH ASSOCIATED GRAVESITE.** The headstones must be carefully excavated and stored on the gravesite without coming into contact with the ground. DO NOT place on dirt piles or mud; they should be carefully placed on each associated gravesite and protected in such a way as to prevent any soiling, and be out of the way of any other work. The headstones must be laid with front inscription side up while the headstone is lying horizontally. The headstone must be protected from direct ground contact while lying horizontally. The protection method must be as approved by the COR and must be free of deterioration in weather. An approved method is to support each headstone with two wood 4x4's. Alternate methods can be approved. Cardboard must not be used. Wood or other suitable appropriate and attractive material must be used to keep the headstones from contact with the soil while lying horizontally during the construction period. This also shows respect towards the families visiting the gravesites and the remains that are buried.

### C. UPRIGHT HEADSTONE SETTING

1. After headstones are removed from their sockets, headstone sockets must be re-dug in accordance with the approved Headstone Realignment Plan providing three (3) inch clearance at the front, back, sides and underneath the headstone, for total socket dimensions of ten inches (10") by nineteen inches (19"). Headstone sockets must be dug to the required depth to meet the specified height and alignment requirements and finish grades established by the approved Grading Plan. All headstone sockets must be clean and free of loose dirt and debris prior to adding crushed aggregate base.
2. Crushed aggregate base must be moisture conditioned within  $\pm 2\%$  of the optimum moisture content and added in 1-1/2" lifts and compacted to 90% to 95% relative density per ASTM D 1557 in the bottom of the dug sockets to provide a minimum three inch (3") footing under the base of the headstone prior to the re-installation of the headstones. The actual measurement will vary depending on the how much the headstone needs to be raised or lowered and must be approved by the COR. (See Exhibit 'B' – Headstone Setting Details.)
  - a. Tamping Tools: Must be approved by COR prior to use. Tamping tools must have sufficient impact area and weight to achieve 90% to 95% compaction of the aggregate. Tamping tools having wood handles and those with insufficient mass are not acceptable.
3. The headstone must be carefully lowered into the socket onto the compacted aggregate base. During the alignment process crushed aggregate backfill must be placed around all four sides of the headstones in 1-1/2" lifts and compacted to 90% to 95% relative density per ASTM D 1557, providing a minimum thirteen inches (13") depth of crushed aggregate backfill along the sides of the headstone as measured from the bottom of the headstone. After establishment of subgrades per the approved Grading Plan there will be a minimum of four inches (4") of space remaining from the top of the compacted aggregate backfill to the ground surface to allow for placement of 2-inches of imported topsoil and turfgrass sod.
4. Existing headstone sockets that exceed the correct depth requirements must be filled with sufficient crushed aggregate base and compacted to 90% to 95% relative density per ASTM D 1557 to achieve the correct final depth. All headstone sockets must be clean and free of loose dirt and debris prior to adding crushed aggregate base. After filling in the deep sockets in this manner, proceed with headstone realignment.

5. In cases where no portion of the existing headstone sockets can be reused due to a significant realignment shift, apply the following procedures:
  - a. After removal of the headstones, backfill and compact the existing socket prior to digging the new socket. Backfill and compact the existing socket with suitable soil or other COR-approved material to 90% to 95% relative density per ASTM D 1557 in 1-1/2" inch lifts making allowance for placement of 2-inches of imported topsoil and sod.
6. The completed raised and realigned headstones must be anchored firmly in place such that the headstones are rigid with no give or play in any direction regardless of the soil and terrain conditions. All headstones must be firmly set and anchored in place with no movement from reasonable forces subjected by the COR, (e.g., using one arm to shake).

D. UPRIGHT HEADSTONE RAISE AND REALIGNMENT:

1. Prior to starting work in each burial section the required gravesite row/gravesite spacing layout dimensions must be verified and certified by a licensed surveyor and coordinated with the COR.
  - a. The Contractor must use a licensed surveyor to install temporary control markers to conform to the section layout.
  - b. The Contractor must use a licensed surveyor to accurately and precisely install temporary control pin markers at the ends of each gravesite row in accordance with the approved Gravesite Realignment Plans.
  - c. The Contractor must use a licensed surveyor to stake locations for new permanent gravesite control monuments as indicated in the approved gravesite realignment plans.
  - d. The Contractor must use a licensed surveyor to accurately and precisely set grade stakes as required to establish finish grades in accordance with the approved Grading Plans.
2. Headstones must be accurately and precisely reinstalled on the correct gravesites by utilizing temporary grave plotting maps, existing permanent gravesite control monuments where available, temporary contractor installed gravesite control markers, and temporary contractor installed control pin markers at the ends of each gravesite row. All measurements, setting of string lines, and the layout of rows/columns must be taken from existing or temporary gravesite control markers, and NOT from previously set headstones, unless otherwise directed by the COR.
3. The proposed finish grades on the approved Contractor-provided Grading Plan (including 2-inches of imported topsoil) must be established by setting grade stakes for the entire section to be raised and aligned in all directions by a licensed surveyor for the approval of the COR. The Contractor must interpolate finish grades to allow all of the headstones in the section to be

aligned vertically without having drastic changes in headstone heights (several inches) between the rows and columns of headstones in ALL directions.

4. In irregular terrain where sloping and uneven ground conditions exist, all headstones and markers will be set at the proper height and overall level necessary to provide a smooth, flowing transition between ALL headstones in ALL directions through uneven terrain. The Contractor must take the adjoining sections and/or terrain into consideration when establishing finish grades. IMPORTANT: To meet the requirement of a smooth, flowing transition, the Contractor may be required to set the headstones at a height other than at the required 25 inches above grade with approval by the COR. (See Exhibit 'C' – Upright Headstone Realignment Diagrams.)
5. Keystone Placement and Installation
  - a. The headstone realignment process starts by the selection of specific headstones throughout the gravesite section to be designated as "Keystones" (or "Key" stones) for use throughout the remainder of the realignment process. Keystones are ones that are visibly in line with the majority of the rest of the headstones in the row as identified by the Gravesite Realignment Plans, and are as close as possible to the proper measurements for that section of headstones. Keystones must be chosen and utilized throughout the section at a maximum spacing of every 10th headstone or at a more frequent spacing as necessary in areas of sloped or rolling terrain at crests and valleys in order to achieve a smooth, flowing transition between the rows and columns. (See Exhibit 'C'.)
  - b. Each of the Keystones must be permanently set in accordance with the requirements of "Upright Headstone Setting" article at the proper location and at the proper height using the approved Gravesite Realignment Maps and approved Grading Plans. The Keystones must be laid out to the correct dimensions to match the required measurements for the section such that the rows and columns are evenly spaced. The vertical height of the Keystones shall be set 25 inches above the proposed finish grades on the approved Grading Plan throughout the entire section in order to achieve a smooth, flowing transition between all the rows and columns. (See Exhibit 'C'.)
  - c. All Keystones must be vertically plumbed and aligned by leveling the front, back, and sides while raising or lowering the headstone as necessary to achieve a height of 25 inches above the proposed finish grades. IMPORTANT: To meet the requirement of a smooth, flowing transition, the Contractor may be required to set the Keystones at a height other than at the required 25 inches above finished grade with approval by the COR.

- d. The COR must approve of the quantity, location, height, and installation of the Keystones prior to the Contractor continuing with the realignment process.
6. The next step in the headstone realignment process is the Contractor selects a row and a column of headstones at the center or close to the center of the section to be realigned with approval from the COR. (See Exhibit 'C'.)
7. Install heavy string lines along the front, sides, tops, and transversely of all applicable Keystones. The string lines must be laid out to the correct dimensions as determined by Keystones and the approved Gravesite Realignment Plans such that the rows and columns are evenly spaced. These lines are marked with the proper measurements for the section being aligned to show distance between headstones. All remaining headstones in the row are then aligned along the strings front to back and side-to-side even with the measured marks on the line. All headstones must be vertically plumbed and aligned by leveling the front, back, and sides while raising or lowering the headstone as required to achieve a height of 25 inches above the proposed finish grades. Maximum vertical, lateral, and transverse tolerance of any headstone off the alignment string lines and/or marks must be 1/8", or less. IMPORTANT: To meet the requirement of a smooth, flowing transition, the Contractor may be required to set the headstone at a height other than at the required 25 inches above grade with approval by the COR. These measurements must be adhered to as closely as possible and may be adjusted to suit unique site conditions with approval by the COR. (See Exhibit 'C'.)
8. The first row of reset/realigned headstones in each burial section must be inspected by COR for appearance, spacing, depth, alignment, plumb, height, accuracy, and smoothness in grade transition. No further setting of headstones shall be done until this first row has been inspected and accepted by the COR. It is the Contractors responsibility to notify the COR 24 hours in advance of when each of these inspections will be needed.
9. All remaining headstones in all rows, columns, and transverse directions are then raised and re-aligned in accordance with above. The installation, adjusting, and setting of the remaining headstones must meet the requirements of the "Headstone Setting" article. (See Exhibit 'C'.)
10. Place and compact aggregate backfill as indicated in "Headstone Setting" article between the headstone and the earthen socket to secure the headstone in alignment. (See Upright Headstone Installation Detail on Exhibit 'B' – Headstone Setting Details.) The alignment of the headstones should be checked frequently during this process because the tamping may move the headstone out of level or off the mark on the line. Maximum

vertical, lateral, and transverse tolerance of any headstone off the alignment string lines and/or marks must be 1/8", or less.

11. Headstones must be installed to within one degree (1°) of fully vertical when measured on front and sides using a digital level.
12. The measurements between rows or columns of headstones and between the headstones within each row or column may differ from one gravesite section to the next. This condition is to be discussed with COR where this is found to occur. These measurements must be adhered to as closely as possible. Some variances may be allowed by the COR in order to keep a uniform appearance of headstones being aligned from one section to the other.
13. Any upright headstones broken or damaged by the Contractor must be reported to the COR or the Cemetery Director by close of business each working day in order that the grave can be properly marked. The Contractor must be responsible for the cost of the headstone replacement. The cost of the replacement headstones will be deducted from Contractor's invoice(s). All headstone replacements must be coordinated with the COR. Any existing permanent gravesite control monuments, temporary Contractor-installed control markers, grid or sectional monuments that are disturbed, displaced, or broken must be replaced and properly reset by a licensed surveyor at the Contractor's expense. Curbs, roads, walks, turf, trees, utilities, etc. existing above and below the ground that are damaged or disturbed by the Contractor during performance of contract work must be repaired at the Contractor's expense at no additional cost to the Government. The requirements of this paragraph must be completed by the Contractor within fourteen (14) workdays, unless otherwise agreed to by the COR.
14. Government furnished headstones that are permanently removed from a grave for any purpose must be transported by the Contractor to an approved location on cemetery grounds and destroyed in any convenient manner, ensuring that the replaced headstone cannot be used for private, personal, or commercial activities.
15. Contractor must inspect and measure each headstone when removed and document any headstones greater in length than the standard 42 inches. At the contractor's option and at no additional cost to the Government, the Contractor can either adjust the depth of the socket to accommodate the longer headstone(s) or cut the headstone to the standard length of 42". Cutting to size will be accomplished locally.

16. Contractor must inspect each headstone when removed and document any damage or physical defect in the stone. Report findings to the COR for direction prior to resetting the headstone.

### **3.2 FLAT MARKER HEADSTONE HANDLING, RESETTING, REALIGNMENT AND BACKFILL REQUIREMENTS:**

- A. Overview: This subsection outlines Flat Marker Headstone Setting, Alignment and Backfill requirements. Setting, Alignment, and Backfill services consist of extracting Flat Markers, constructing the Flat Marker support system, resetting, aligning, backfilling and tamping/compacting flat marker headstones to provide vertical and horizontal alignment and plumb, as well as inventorying and assuring accurate placement of headstones onto gravesites. The Contractor must provide all supervision, professional advice/guidance, labor, parts, materials, equipment, and personnel necessary to provide the services defined herein.
- B. Marker Locations: Verify with the COR the exact marker locations where Flat Marker support system, resetting, realignment, and backfill shall occur.
- C. Phasing and Work Sequencing: All work must be accomplished in a sequential manner, with work limited to no more than 2000 Marker gravesites at any given time in order to minimize overall disruption to the cemetery. The Contractor must submit a proposed "Project Work Schedule Sequence" to the COR for review and approval prior to start of project.
- D. There are two types of Flat Markers in the work area. Flat Granite Markers (G1) are approximately 24-inches long by 12- inches wide by 4 inches thick, weighing approximately 130 pounds. Thin Flat Granite Markers (J1) are approximately 24-inches long by 12- inches wide by 3 inches thick, weighing approximately 100 pounds; Small Flat Granite Markers (L1) are approximately 18-inches long by 12-inches wide by 3-inches thick, weighing approximately 50 pounds. The term "Flat Marker Headstone" may be used interchangeably with the terms "Flat Marker(s)".
- E. Flat Marker Headstone Record Keeping and Handling
  1. Handling of Flat Markers: Contractor must create an as-built map showing specific location of each individual Flat Markers by actual number and name(s) in order to facilitate precise Flat Markers reinstallation.
  2. The measurements between rows of flat markers, and flat marker within each row may differ from one burial section to the next. Discuss this issue with the COR where this is found to occur. These measurements must be adhered to as closely as possible. Some variances may be allowed in order to keep a uniform appearance of flat markers being aligned from one section to the other. Markers must be accurately and precisely reinstalled on the correct

gravesites with the use of temporary grave plotting maps, existing permanent control markers where available, with temporary contractor installed control markers accurately and precisely installed at the ends of each gravesite row.

F. Flat Marker Alignment, General

1. All measurements and string line set-ups must be taken from established section control markers and/or from temporary control markers installed by the contractor around the section and at the end of each grave row, not from previously set flat markers, unless otherwise directed by the COR. All measurements must be made in ascending grave number order. (See Exhibit 'D' – Flat Marker Realignment Diagrams.)
2. All markers in flat terrain must be leveled laterally and transversely using a bubble level. In rolling terrain where sloping ground conditions exist, all flat markers must be set with top surface at proper heights and levels to provide a uniform flowing transition through the rolling terrain.
3. Markers must be accurately and precisely reinstalled on the correct gravesites with the utilization of temporary grave plotting maps, existing permanent control markers where available and temporary contractor installed control markers accurately and precisely installed at the ends of each gravesite row.
4. Flat markers must be set in a line laterally, transversely, and diagonally with flat markers of other graves, and with flat markers in adjacent sections.
5. After all work has been completed, the flat markers in all soil and terrain conditions must be held firmly in place by the compacted topsoil and aggregate setting bed so that the flat markers are rigid with no give, play, or movement when subjected to forces by the COR.

G. Flat Marker Headstone Setting, and Alignment onto a Headstone Support System must be accomplished as follows:

1. The Contractor must pull the flat markers from the flat marker sockets and carefully store the markers on each associated gravesite.
2. (See Flat Marker Installation Details on Exhibit 'B' – Headstone Setting Details.)
3. Excavate a continuous trench under the marker support system. Compact the subgrade to 95 percent maximum dry density in accordance with ASTM D698.
4. Place and compact 3/4" minus crushed aggregate base under the plastic geogrid at marker locations. Compact to 92 to 95 percent maximum dry

density in accordance with ASTM D1557 and establish proper grade for the placement of the plastic headstone support geogrid and the subsequent setting and alignment of the flat markers.

5. Place and compact amended topsoil underneath the plastic support geogrid where sod is to be planted. Compact to 90 to 92 percent maximum dry density in accordance with ASTM D698 and establish proper grade for the placement of the plastic headstone support geogrid.
6. Install continuous strip of plastic headstone support geogrid along the entire length of each flat marker row beginning at the edge of the first flat marker and continuing until reaching the far edge of the last flat marker in the same row. Connect plastic support grids together using snap on connectors in accordance with the manufacturer's printed instructions.
7. Fill voids in geogrid beneath the flat markers with 1/4-inch minus crushed aggregate levelling sand and compact with flat plate compactor.
8. Set Flat markers onto the plastic geogrid. Flat markers must be set to a depth so that the top surface of the flat marker is 1-inch above final grade in accordance with Fort Snelling National Cemetery Master Plan.
9. Place and compact additional 1/4-inch minus crushed aggregate levelling sand to level flat markers as required, maximum 1-inch depth.

#### H. Aligning Flat Markers onto the Headstone Support System

1. The realign crew starts on a row of flat markers close to the center of the section to be realigned. Markers are chosen to be "key" stones. These keystones are ones that are visibly in line with the majority of the rest of the markers in the row, and are as close as possible to the proper measurements for that section of markers. "Key" stones must be chosen and utilized at a maximum of every 10th marker (or more frequently in areas of rolling terrain). These keystones are then aligned by leveling front and back and side-to-side, and raised or lowered to a final elevation relative to the topsoil level as determined by the COR.
2. At the start of work in each burial section, the first row of reset/realigned flat markers in each burial section will be inspected by COR for appearance, spacing, depth, alignment, plumbness, height, accuracy, and smoothness in grade transition. No further setting of flat markers must be done until this first row has been inspected and accepted by the COR. It is the Contractors responsibility to notify the COR 24 hours in advance of when each of these inspections will be needed.

3. Heavy strings or lines must run along the side and top edges of the keystones. All line set-ups must be secured to retain proper alignment even in windy conditions. These lines are marked with the proper measurements- (size of section) for the section being aligned. All remaining flat markers in the row are then aligned along the strings front to back and side-to-side even with the measured marks on the line. Each flat marker must be adjusted front to back and side-to-side keeping the back of the flat marker along the string and the side of the flat marker along the measured mark. Maximum vertical, lateral, and transverse tolerance of any flat marker off the alignment string lines and/or marks must be 1/8-inch, or less.
  4. Each flat marker must be raised or lowered as necessary to be 1 inch above topsoil finished grade subject to the top string slopes required for the markers to flow with the sloping terrain. The flat marker must then be aligned along the string front to back and side-to-side even with the measured mark on the line. In flat terrain, the flat marker must be leveled and plumbed front to back and side-to-side keeping the back of marker along the string and the side of the marker along the measured mark.
  5. Adjustments to marker height and alignment must be made by placing and compacting 1/4-inch minus crushed aggregate levelling sand on top of the marker support geogrid.
  6. The alignment of the flat markers must be checked frequently during this process to ensure that the marker is not out of level or off the mark on the line. This process of raising and realigning is repeated for each row of markers. Flat markers in all completed work areas must be firmly set and anchored in place with no movement from forces subjected by the COR.
- I. Backfilling
1. Backfill amended topsoil into the remaining voids of the geogrid located between the flat markers and firmly compact this material into place.
  2. Backfill remaining gravesites areas around and between all flat markers from the subgrade to top of finish grade surface with 4 inches minimum of topsoil compacted to 85 percent maximum dry density in accordance with ASTM D-698 so that no settlement shall occur. Footprints must not be visible on the compacted topsoil when walked upon.

### **3.3 RECORD KEEPING AND REPORTING:**

- A. The Contractor's Superintendent (or their designee) and COR (or COR's Representative(s)) must conduct a joint gravesite inspection and verification prior to the handling of any flat marker headstones to document the location and accuracy of all existing conditions. The COR will provide a printout of the burial records (BOSS) for this purpose with the following data fields. Section, Row,

- First and last name of the deceased. Upon completion of this joint survey, both the contractor's Superintendent and COR must certify the accuracy, as noted and initialed, by signing the burial record printout.
- B. Upon completion of the resetting and realignment of the headstones and markers, contractor's Superintendent and COR re-inspect and verify the accuracy of the placement of headstone and marker on the correct gravesite. Upon completion of this second joint survey, both the contractor's Superintendent and COR must certify the accuracy by re-signing the burial record printout.
  - C. Record Keeping and Reporting: The Contractor's Project Manager must provide the COR(s) with weekly written accurate reports detailing the section and range of graves in that specific week. These lists will be provided on not less than a weekly basis, and must identify all of the above work that took place within the previous seven calendar days. Upon request, the Contractor's Project Manager must also provide a plan of action for the upcoming week, regarding headstone/marker locations and specific areas where work is being scheduled to occur.

### **3.4 HEADSTONE CLEANING:**

- A. Headstone Cleaning, General: The Contractor must provide all labor, parts, equipment, supplies, transportation and personnel required to provide headstone/marker cleaning services.
- B. Definition: "Clean" under this contract means: The headstones/markers shall contain NO discoloration, environmental deposits, mold, mildew, moss, algae, lichen, dirt/mud, grass clippings, grass stains, tree sap, bird droppings, or other foreign stains and markings.
- C. Contractor Duties and Responsibilities:
  - 1. All equipment and supplies maintained and operated by the Contractor must be consistent and fully compliant with all applicable Federal, State, and County laws, ordinances and regulations and meet State inspection, safety, licensing, registration, and insurance requirements.
- D. Test-Cleaning Flat Marker Headstones
  - 1. The COR shall select five (5) badly-stained marble upright headstones and five (5) badly-stained granite flat markers for test cleaning in accordance with the manufacturer's printed application instructions. Use dilution rates of cleaning solution as recommended by the manufacturer for granite and marble. Vary level of agitation, contact time, and rinsing for each of the five (5) test markers to determine optimum cleaning method. Allow a minimum of 7 days drying time before final inspection. Test markers shall serve as the

performance standard and remain available for comparison during the cleaning process.

2. Each test marker must be carefully labeled, charted, and photographed. Approved test markers will become a part of the Work, and serve as the quality standard for similar type work on this project. Re-clean test markers not meeting quality standard.
3. Contractor must prepare a written report detailing results of testing including description of methods employed, materials, concentration of cleaner, dwell times and other elements of test procedures as defined above.
4. New turf sod, if installed at the time of the cleaning, must show no negative effects as a result of the cleaning process. Sod damage is unacceptable. Additional actions such as rinsing with water and/or masking of turf during application of cleaner and pressure-washing will be required at no additional cost to the Government.

E. Cleaning of Flat Marker Headstones:

1. All temporary markers, floral, commemorative, or other types of decorations (arrangements) causing interference with the cleaning of headstones must be carefully and orderly moved from and, upon completion of the cleaning of the markers moved back to the gravesite by the Contractor.
2. Cleaning techniques must demonstrate a clear understanding of, and the sensitivity to, such environmental issues as ground water contamination, wetlands, etc., and must be consistent and fully compliant with all applicable Federal, State, and County laws, ordinances and regulations. Clean water must be used to clean headstones. Cleaning techniques with water shall include high pressure spraying, hand scrubbing, and rinsing. When water under pressure is used, such pressure must not be greater than 500 psi. Excessive staining and/or discoloration may be removed with pressurized water and a stiff nylon brush (no wire brushes, bleach, or abrasive cleaners shall be permitted) followed by rinsing with clear water. If water used in cleaning should soften the soil around the footing of the headstone so that the headstone is loosened, care must be taken not to tip the headstone out of plumb or alignment. Care must be taken to protect the turf area from damage. Any turf damaged by the Contractor must be restored at Contractor's expense. Headstones must be set and anchored firmly in place with no movement from forces subjected by the COR or inspector after cleaning has been completed.
3. Applying the cleaner: Apply the cleaner according to the manufacturer's printed instructions. Mask the surrounding turf to minimize contact with the turf. Do not allow the cleaner to dry on the turf. Do not use cleaner product in a greater strength than the manufacturer recommends. Evenly apply the

cleaner with a sprayer to saturate the surface. Always keep the stone wet during cleaning and thoroughly rinse afterwards.

4. Agitating the surface: Agitate the surface gently in a circular motion using a soft bristle brush. Work in small areas, starting from the bottom and moving toward the top of the headstone. Agitation will loosen soiling from the surface of the stone. Soft bristle brushes are required when cleaning stones. They can have natural or synthetic bristles. Vegetable brushes or soft grooming brushes for large animals are a few that can be found in chain or farm supply stores. All rough or metal edges must be covered with tape to reduce the chance of scratching the stone.
5. Rinsing the stone: Rinse after cleaning each area and to thoroughly rinse the stone at the end to make sure that no cleaner is left behind. A typical cleaning regime may include a three-person team. The first person sprays the stone surface with the biocidal cleaner. After the appropriate dwell time, a second person gently agitates the cleaner on the stone surface with a soft bristle brush, then a third person rinses the stone with clean tap water.
6. Pressure wash markers from a distance no closer than 12 inches and with a pressure of 500 psi or less. Some stones may not be able to tolerate these conditions depending on their condition. A test patch in a small unobtrusive area on the headstone is recommended prior to cleaning.
7. Contractor must submit the required Safety Data Sheets (SDS) to the COR prior to use of any chemicals including but not limited to 'D/2' and 'Weatherzyme'.
8. Should any headstones that have been cleaned but become marked, discolored, dirt covered, or muddied due to subsequent contract work including but not limited to turf re-establishment issues must be re-cleaned at no additional cost to the Government. Headstones that become discolored, dirt covered, or muddied, etc. after initial cleaning has been completed, but prior to acceptance, must be re-cleaned at no additional cost to the Government.
9. The Contractor's Superintendent must provide the COR(s) with weekly written accurate reports identifying the following: All headstones cleaned must be recorded per gravesite location. These lists will be provided on not less than a weekly basis, and must identify all of the above work that took place within the previous seven calendar days.

### **3.5 DAMAGE AND WARRANTY:**

- A. Any headstones broken or damaged by the Contractor must be reported to the Cemetery Administrator or COR by close of business each working day in order

- that the grave can be properly marked. The Contractor must be responsible for the cost of the replacement if it is determined to be due to contractor not following proper procedures. The cost to replace contractor damaged upright headstones and flat markers will be deducted from the contract price at the rate of \$300.00 per headstone. Any grid or sectional monuments disturbed, displaced or broken must be replaced by the Contractor at his/her cost.
- B. Any grid or sectional monuments disturbed, must be properly reset by a licensed land surveyor at Contractor's expense at no additional cost to the Government. Curbs, roads, walks, turf, trees, utilities existing above and below the ground that are damaged or disturbed by the contractor during performance of contract work must be repaired at the expense of the contractor. Repairs to the above must be corrected by the Contractor within fourteen (14) workdays, unless otherwise agreed to with the COR.
- C. The Contractor is to warranty all work for a period of one (1) year after Final Acceptance by the Government. Workmanship pertaining to headstone adjustment, realignment, resetting, and backfilling that does not meet the specified requirements (including specified tolerance requirements) before the end of this warranty period is to be reworked, adjusted, and corrected by Contractor at no additional cost to the Government. Work performed under the warranty must be corrected within ten (10) workdays from receipt of notification or as directed by the COR.

## **END OF SECTION**

### **SECTION 32 92 24 - TURF RENOVATION**

#### **PART 1 - GENERAL**

##### **1.1 TURF RENOVATION WITH SOD:**

- A. General: Existing turf and weeds must be killed, removed and replaced with new sod as shown on the drawings. The area must be expertly graded and re-sodded resulting in a uniform stand of turf, high quality in appearance: Tops of upright headstone shall be 25 inches above finished grade measured to the top of the arc; tops of all flat marker headstones shall be 1-inch above the soil; or to a measurement as directed by the COR.
- B. Erosion and Sediment Control: Contractor must install and maintain erosion control material/methods to meet Federal, State, and local requirements. Copies of these requirements may be reviewed by contacting the COR(s). As necessary, the Contractor must install measures including, but not limited to, sedimentation fences and protection of storm drain inlets. Contractor is solely responsible to meet all Federal, State and local requirements for air quality, dust

and particulate matter control. The Contractor is fully responsible for the control of any dust carried by the wind and will take the appropriate preventative measures.

C. Topsoil Placement and Compaction: See “SECTION 31 00 00 - EARTHWORK, Paragraph 3.8 TOPSOIL PLACEMENT”.

D. Agricultural Soil Testing:

1. Imported Topsoil: Soil samples must be collected from each source of imported topsoil intended for use on the project and be submitted to an independent, certified soil-testing laboratory to determine fertility requirements. Soil testing laboratories affiliated with a vendor or supplier will not be accepted. Contractor must provide a copy of each soil test results and recommendations to the COR and Midwest District Agronomist for review and approval. Soil test reports must indicate that the proposed imported topsoil(s) meets the specification for Topsoil as defined herein. Amend topsoil not meeting the pH, organic matter content, and fertility range specified by the addition of pH adjusters, organic soil amendments, and fertilizers.
2. On-Site Existing Soil: Representative soil samples must be collected from a minimum of six (6) regions within the cemetery and be submitted to a certified agricultural soil-testing laboratory four to eight weeks prior to the intended re-sodding operation to determine fertility and pH adjustment requirements. Six (6) soil test reports are required, one for each of the regions defined below. Soil test reports must be labeled with the cemetery sections that are included in each region. Contractor must provide a copy of each soil test results and recommendations to the COR and Midwest District Agronomist for review and approval. As determined by the soil test, Elemental Sulfur and Gypsum may be required to adjust the soil pH and improve permeability, organic soil amendments to increase soil organic matter content, fertilizers to correct soil nutrient deficiencies, as well as other soil amendments to meet the specifications.
3. Soil extractions must be performed in accordance with standard guidelines for agricultural soil sampling and under the observation of the COR.
  - a. A representative sample should consist of a composite of 15 to 20 sub-samples taken from the region where a single soil test is required. Equipment for sampling may be a spade, shovel, auger or soil sampling tube. Sub-samples should be extracted evenly throughout the region.
  - b. Collect sub-samples from the anticipated root zone, (1 to 6 inches for turf), place in bucket, and mix thoroughly. Remove roots and leaves from the sample. Fill a clear, zip-top plastic bag with 1-quart of the soil

- and label each composite sample with Cemetery Name, Region number and Sections numbers.
- c. Repeat for each of the remaining soil testing regions.
  - d. Attach a copy of this article to the work order so that the soils lab knows which soil analyses and recommendations are required.
4. Provide soil analyses in the form of a graphical report showing:
- a. Soil composition (USDA particle size analysis indicating percentages of sand, silt and clay).
  - b. Percent organic matter.
  - c. Soil pH.
  - d. Macro and micronutrient fertility tests as determined by pH.
  - e. Salinity (ECe).
  - f. Nitrate nitrogen, ammonium nitrogen, phosphate phosphorous, potassium, calcium, magnesium, soluble copper, zinc, manganese, iron saturation extract, boron and sodium analyses.
  - g. Cation exchange capacity (CEC).
  - h. Sodium Adsorption Ratio (SAR).
  - i. Soil infiltration rate.
5. Report suitability of tested soils for plant growth: Based upon the test results, the laboratory must provide recommendations for fertilizers and soil amendments to be applied to the soils for the establishment and maintenance of healthy turf planted from sod.
- a. State recommendations for macro and micronutrients in pounds per 1,000 square foot, or tons per acre, as necessary to correct soil nutrient deficiencies.
  - b. State recommendations for soil amendments to be added in pounds per 1,000 square foot, or tons per acre to adjust soil pH to within the range of 6.5–7.5. (Limit elemental sulfur to 50 lbs/1000 sq-ft per application.)
  - c. State recommendations for soil amendments to be added in pounds per 1,000 square foot, or tons per acre to improve soil permeability and increase soil infiltration rates.
  - d. State recommendations for compost to be added in cubic yards per 1000 square feet to adjust soil organic matter content of the soil to a minimum of 4 percent and maximum of 6 percent.

## **PART 2 – PRODUCTS**

**2.1 COMPOST:** See “SECTION 31 00 00 - EARTHWORK; article 2.1.I.”

**2.2 STARTER FERTILIZER:** Commercial, general-purpose turf starter fertilizer with a typical N:P:K analyses of: 16-24-10 or 16-16-16.

**2.3 MULCH:**

- A. Douglas Fir Bark, sized 5/8-inch minus with 100% passing 1-inch screen, to match existing bark mulch. Mulch material must be aesthetically pleasing, bark mulch, dark brown, free of any weeds and pests.
- B. Submit a 1/2 cubic foot sample to the COR for approval, prior to installation.

#### **2.4 SOD:**

- A. Sod must be nursery grown as classified in the TPI Guideline Specifications to Turfgrass Sodding, produced from Blue Tag certified seed and State of Minnesota certified. The composition of the grass species in the sod shall be a mix of at least three (3) seed cultivars. Each grass species must be a regionally adapted cultivar. The sod must be free from weeds, pests and diseases. Seed species must have achieved a NTEP average rating of 5.5 or greater out of 9. Submit sod grower's product data sheets to COR for approval prior to ordering sod.

#### **2.5 TURF MAINTENANCE PRODUCTS:**

- A. Turfgrass fertilizers based on recommendations of soils reports. commercial grade and uniform in composition, free-flowing granular complete analysis fertilizer containing a minimum of 20% by weight of nitrogen, 5% phosphoric acid and 15% potash with a minimum of 50% of the nitrogen from a controlled release source such as sulfur-coated urea.
- B. Soil conditioners, application rate and frequency based on recommendations of soils reports.
- C. Post-emergent selective herbicides for control of broadleaf and grass weeds, sedges, and other weeds in cool season turf.

### **PART 3 – EXECUTION**

#### **3.1 EROSION AND SEDIMENT CONTROL:**

- A. Contractor must Install and maintain erosion control material/methods to meet Federal, State, and local requirements. Copies of these requirements may be reviewed by contacting the COR(s). As necessary, the Contractor must install measures including, but not limited to, sedimentation fences and protection of storm drain inlets.
- B. Contractor is solely responsible to meet with all Federal, State and local requirements for air quality, dust and particulate matter control. The Contractor is fully responsible for the control of any dust carried by the wind and will take the appropriate preventative measures.

### **3.2 IRRIGATION SYSTEMS:**

- A. Testing and Demonstration: Cemetery personnel will test and demonstrate to the Contractor all irrigation systems and hardware prior to the start of work in any area. This inspection must be documented in the weekly work report. Note any system deficiencies in the report.
- B. Preparation: Prior to commencing any work, the Contractor will remove the irrigation heads, cap risers, and turn the heads over to the cemetery. Flag location of heads to prevent damage to irrigation piping during earthwork activities. Maintain flags until after installation of sod.
- C. Reinstallation and Adjustments:
  - 1. After headstones are reset and aligned and topsoil has been placed and compacted, uncap the existing sprinkler risers, flush irrigation lines and install the new irrigation heads in preparation for weed control measures.
  - 2. Set irrigation heads to proper height relative to finish grade for the sod installed as directed by the COR.
  - 3. Coordinate installation of irrigation heads within the turf paving system with the installer.
  - 4. Test irrigation system for leaks and proper functioning and make any necessary repairs.
  - 5. Make any nozzle changes as directed by the COR to achieve 100 percent head-to-head coverage. Make adjustments to arc and radius of the spray pattern as directed by the COR.
  - 6. Prior to the installation of sod, Contractor must ensure that all irrigation heads, valve boxes, and related components are 1 inch above finished grade, aligned correctly and in proper working order.
- D. Testing and Repair
  - 1. Any damage to the irrigation system caused by the Contractor must be repaired by the Contractor prior to sod installation at no additional cost to the Government.
  - 2. Prior to installing sod, demonstrate to the COR that the irrigation zone to receive sod is fully functional from the irrigation controller. Sod installation must be sequenced to conform to existing irrigation zones, such that sod is laid in areas that can be completely irrigated by the existing irrigation system without causing disruption to work in other areas.

### **3.3 VEGETATION REMOVAL:**

- A. General: Remove turfgrass sod, weeds and their roots from within limits of work area. Cut a clean line between sod to be removed and sod in adjacent cemetery sections to be retained.

- B. Protection of Headstones: Ensure that the work involved for vegetation removal does not cause the headstones to be shifted out of alignment or moved from their sockets. Any misalignment of headstones must be corrected by the Contractor at no additional cost to the Government.
- C. Herbicide Application: Apply herbicide in a minimum of TWO APPLICATIONS in the sequence indicated below.
  - 1. First application: Do not turn off the irrigation system until just before the application of the herbicides. This will allow the turf to absorb the maximum amount of herbicide. Tank-mix herbicides following precautions of herbicide manufacturers.
  - 2. After the preparatory mowings, while the turf is still actively growing, the entire vegetated area must be sprayed with non-selective and selective herbicides, taking care to avoid drift of herbicides onto adjacent vegetation to be protected.
  - 3. Application rate shall be at the maximum label recommended rate for the complete kill of the existing turfgrass, *Poa annua*, sedges and other weeds.
  - 4. Follow product label instructions for elapsed time before moving to the next procedure to allow the herbicides to fully affect the plant material.
  - 5. Power rake or verticut the entire area to loosen residual plant debris, taking care not to disturb the headstone support systems where noted to be protected. Remove plant debris and dispose of legally off site.
  - 6. The area shall then be irrigated for a minimum of 14 days to encourage growth of any remaining plant material and germination of weed seed lying dormant in the topsoil.
  - 7. Second Application: When there is sufficient re-growth of vegetation as determined by the COR/Cemetery Director, the area will be retreated with non-selective and selective herbicides, again following the product label directions for time to elapse for the herbicides to fully affect the plants, before the start of soil preparation activities.
  - 8. Once again, Power rake or verticut the entire area to loosen residual plant debris as indicated above. Remove plant debris and dispose of legally off site.
  - 9. The Contractor must repeat this operation until the existing turf, *Poa annua*, sedges and other weeds have been eliminated, as determined by the COR/Cemetery Director.

### **3.4 IMPORTED TOPSOIL:**

- A. Prior to placing imported topsoil, broadcast mycorrhizae onto the surface of the scarified subgrade. (See SECTION 32 00 00 – EARTHWORK). Place and compact 2-inches of imported topsoil to depths indicated in the Drawings to establish uniform finish grades as indicated in the approved Contractor-provided grading plan: Finish grades shall be 1-inch below flat

markers; 3/4 to 1-inch (or a depth equal to the thickness of the sod root mass) below tops of curbs, concrete pads, vaults, valve boxes, etc. as directed by the COR/Cemetery Director.

### **3.5 AMEND TOPSOIL:**

- B. Add preplant fertilizers, compost, elemental sulfur, gypsum and any other soil amendments recommended by the soils report to correct any deficiencies in the imported topsoils and to meet the requirements of these specifications.
- C. Till the soil to a depth of 4 inches to uniformly incorporate soil amendments and imported topsoil into the native topsoil to produce amended topsoil. Areas within the drip lines of existing trees shall not be tilled at a depth greater than 3 inches or backfilled not to exceed 2 inches from existing grades, or as approved by the COR. Any damage to tree roots must be avoided. If tree roots within the drip-line of a standing tree that are 1 inch in diameter or greater are encountered, the Contractor must request an inspection by the COR. The COR will inspect and determine if any roots will be pruned. The preference will be to relocate the roots into a deeper soil profile to avoid cutting of roots. Contractor must protect all trees, shrubs and plants. Plant material damaged, weakened or killed during the renovation must be replaced with its equivalent. Exceptions must be approved by the COR.
- D. Amended Topsoil (post-amended in-place soil testing): Following the incorporation of soil amendments, submit samples of amended topsoil for testing and verification that the soils meet the specifications and have no significant nutrient deficiencies. Contractor shall deliver samples to testing laboratory and shall have the test reports sent directly to the CO/COR. Tests shall be as conducted as indicated in paragraph 1.1 D.2 of this Section. No sodding shall occur until the COR has approved the amended topsoil.

### **3.6 FINE GRADING AND COMPACTION:**

- A. General: The area will be graded and compacted to provide a smooth and flowing transition through uneven terrain and to avoid any appearance of layering of soil. Contractor shall use a laser level or similar device to maintain grade lines.
- B. Compaction: Restore the soil to an even condition before laying sod by rolling with a turfgrass roller well-suited to the soil being compacted. Do not operate mechanized vibratory compaction equipment, which may damage existing graveliners or caskets. The Contractor must not operate trucks, tractors, and other heavy equipment in excess of three (3) tons on any turf area except when authorized in writing by the COR/Cemetery

Director. Finish grades must be thoroughly compacted to 85 to 90 percent dry density per ASTM D-698, to prevent further subsidence, but the soil must not be compacted to the degree that it prohibits the growth of healthy turf. The compacted soil must not show evidence of depressions when walked upon.

- C. Contractor must establish lines and grades for approval by COR/Cemetery Director or designee. Grades must conform to elevations to provide a smooth transition at curbs, trees, planters and uninterrupted drainage flow into existing drains, and to prevent any “scalping” of the new turfgrass when mowed.
- D. Contractor must request COR/Cemetery Director’s approval of lines and grades. Contractor will be responsible for any additional cut and/or fill required to ensure that the site is graded to conform to elevations as determined by the COR/Cemetery Director.

### **3.7 SOD BED PREPARATION:**

- A. The finished bed must be fine in texture and firmly compacted and free of any stones, plant or other debris greater than 13mm (1/2 inch). All irregularities in the finished surface must be corrected to eliminate depressions and high spots.
- B. Finish grade of soil shall be the depth of sod root mass (approx. 3/4 to 1-inch) below adjacent pavement, and flush with finish grades of adjacent burial sections.
- C. Valve boxes, sprinkler heads and related irrigation components are to be adjusted to 1-inch above finish grade or as directed by the COR.
- D. Complete turf renovation only after areas are brought to the final grade and have been reviewed and approved by the COR.
- E. All finished topsoil areas must be protected from damage from vehicular or pedestrian traffic.

### **3.8 STARTER FERTILIZER:**

- A. Two applications of starter fertilizers are required, one prior to sod placement and another after the first mowing. The fertilizer application shall be high in phosphorous and provide no greater than 1 pound of Nitrogen and at least 1 pound of Phosphorous per 1,000 square feet.
- B. Use properly calibrated fertilizer spreader equipment to broadcast fertilizers at prescribed application rates.

### **3.9 SOD INSTALLATION:**

- A. Prior to delivery of sod, the COR(s) shall inspect the work area. Any discrepancies in the ground preparation must be corrected prior to the laying of sod in the work area.
- B. Prior to installation of the sod, the COR shall have the right to inspect and to assess the acceptability and quality of the proposed sod. The COR shall have the right to reject poor quality sod before installation.
- C. The sod shall be cut as thinly as possible (3/4 to 1-inch) to allow for faster rooting and must be cut and delivered to the work site the same day of installation. Contractor must make all necessary arrangements to protect delivered sod from excessive drying and wind damage.
- D. Sod must be laid expeditiously without interruption, until the work area is completely sodded.
- E. Stagger transverse joints in sod in a “running bond” pattern so seams in adjacent strips do not align. Sod must be laid with seams tightly matched, and then tamped lightly or rolled to ensure firm contact with the soil and no air gaps.
- F. Sod placement must conform to the existing adjacent grade. On slopes, sod shall be laid perpendicular to the direction of the slope.
- G. Sod must be cut and fitted around all markers, headstones, sprinkler heads, valve boxes, edges of paving/curbs, and other fixed objects.
- H. Sod must be cut around trees to provide location for mulch tree rings. Size of mulch rings shall be determined by the COR/Cemetery Director, but in no case less than 24 inches radius scribed around the tree trunk. For coniferous trees with branching at ground level, the mulch ring shall be cut 12 inches beyond the dripline to allow access for lawn mowers.
- I. Water sod with fine water spray within two hours of planting, or sooner if temperatures are above 80 degrees F. Irrigate to ensure thorough wetting of the soil to a depth of at least 3 inches (75 mm). Sod must be kept continually moist until it is well rooted and able to survive with standard watering.
- J. Immediately after installation of the sod, protect against traffic or other use by erecting barricades, as required, and placing approved signs at appropriate intervals until Final Acceptance.

- K. The contractor must warranty the sod for one (1) year after the date of Final Acceptance of the project.

### **3.10 REQUESTS TO IRRIGATE:**

- A. Requests to irrigate newly placed sod, to include amount and frequency, must be made to and approved by the COR/Cemetery Director, or their designee. Access to and programming of the irrigation remote control valves will be performed by cemetery personnel at the direction of the COR/Cemetery Director or their designee. Contractor will not manually override the electronic irrigation control system unless specifically authorized to do so by the COR/Cemetery Director, or their designee.

### **3.11 MULCH:**

- A. Cut and remove sod in a circle around the trunks of existing trees. Provide a 3-inch deep layer mulch material to tree mulch rings, and other areas surrounding groves of trees where lack of sunlight and/or preponderance of tree roots preclude the growth of turf. The outline and extent of these areas will be determined in the field by the COR at the time of turf area surveying.
- B. Maintain mulch in a uniform depth. Hand weed and/or use appropriate chemical controls for weed control in mulch beds until Final Acceptance.

### **3.12 TURF ESTABLISHMENT PERIOD:**

- A. The Establishment Period for turf shall begin immediately after complete sodding of a section(s) with the approval of the COR or designee, and continue for a minimum of 30 days, at which time the Government will perform the final "turf inspection" and acceptance for that particular section(s).
- B. During the 30 days Turf Establishment Period, the Contractor must:
1. Eradicate all weeds. Fertilize, re-sod, and perform any other operation necessary to promote the growth of uniform, healthy, high quality turf.
  2. Irrigate to keep the sod moist and healthy to promote rooting. Conserve water by scheduling multiple start times to prevent run-off. Adjust sprinkler heads to minimize overspray onto pavement.
  3. Replant areas void of turf 1/2 sq-ft (one-half square foot) and larger in area with matching sod species.
  4. Begin mowing with newly sharpened mower blades when grass is 100 mm (4 inches) high to a height of 65 mm to 75mm (2-1/2 inch to 3

inches) and maintain the turf at that height. Never remove more than 1/3 the leaf surface in a single mowing. Turf is also to be trimmed between the headstone markers to the same height of the surrounding turf without scalping.

### **3.13 CLEAN UP AND REPAIR:**

- A. Upon completion of the work, the Contractor must remove all debris, rubbish, and excess material from Cemetery Property unless noted otherwise.
- B. A temporary storage site shall be provided by the cemetery for short-term storage. Contractor may provide own storage container for tools, equipment, and materials purchased for performance on this contract. Storage container shall not exceed 22 feet in length and can only be placed where approved by COR.
- C. At the end of each day the Contractor must remove all debris from the work site to disposal areas. The Contractor must ensure at all times that rubbish and trash generated by the Contractor is removed from paved areas and is kept clear of vehicular and pedestrian traffic throughout the site. The Contractor is fully responsible for disposal of debris. The Government shall not provide receptacle(s) for disposal of debris related to this contract. All unsuitable/unusable materials are to be disposed of off-site in a legal manner.
- D. Dispose of soils and organic matter, plastic, glass, metal, paper waste and concrete generated by the Contractor off-site in a legal manner.
- E. Contractors must be responsible for cleaning any cemetery structures that are soiled or stained as a result of contractor's performance. The Contractor must wash-down with water all soiled or stained structures and grounds at the end of each workday. No chemicals are to be used during the wash down. The Contractor must bear all costs associated with washing and cleaning. Any such soiled or stained areas must be brought to the immediate attention of the COR(s) prior to washing/cleaning.
- F. In areas where planting and turf work have been completed, clear the area of all debris, spoil piles, and containers. Clear all other paved areas when work in adjacent areas is completed. Remove all debris, rubbish and excess material from the Cemetery, except as permitted herein. Segregate and stockpile excess topsoil and organic soil amendments, and haul to the cemetery spoils area, or as directed by the COR.
- G. The contractor and COR(s) shall inspect the trees and shrubbery of a scheduled turf renovation area prior to the start of the renovation and note

all abnormalities or existing conditions. Another inspection will be conducted after the completion of the renovation and the contractor will make all corrective actions to any noted damage. A licensed arborist will complete all corrective actions to damaged trees at the Contractor's expense.

- H. Where existing or new turf areas have been damaged or scarred during planting and construction operations, the Contractor must restore disturbed area to their original condition.

**END OF SECTION**

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